PARAMETERIZED COMPLEXITY AND GEOMETRIC APPROXIMATION OF RIESZ ENERGY SUBSET SELECTION

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ABSTRACT

Selecting k sites of minimum inverse-square Riesz energy is polynomial on a line but NP-hard in the plane. Adapting the proxy lemma of Ceccarello et al. gives an efficient parameterized approximation scheme (EPAS) for fixed Euclidean dimension and Riesz exponent $s > 0$. We sharpen the resulting bound by averaging the interaction cost of one selected site over feasible replacements among $2k$ farthest-first centers. In the plane at $s = 2$, this improves the representative bound from $O(k^2 \log(2k)/\varepsilon^2)$ for a packing estimate of total energy to $O(k \log(2k)/\varepsilon^2)$. We construct rational planar instances where every optimum has a pair at distance $\Theta(r/\sqrt{\log k})$, matching the separation bound in terms of the last center's insertion distance r. A $(1 + \varepsilon)$ approximation on n rational planar sites takes $O(nk + n \log n) + (C \log(2k)/\varepsilon^2)^k$ operations of polynomial bit cost. The observed count M of representatives gives the search size $\binom{M}{k}$. On a million sites, the exchange grid left 252 five-site subsets for clustered data and about 9.3×10^{26} for uniform data. The argument extends to bounded-doubling metrics and to compact domains with effective geometric routines, including filled simple polygons. At $s = 2$, exact decision on metrics with nonzero distances in $\{1, 2\}$ is W[1]-complete in k. A gap shrinking with k rules out a deterministic EPAS on unrestricted metrics unless $\mathrm{FPT} = \mathsf{W}[1]$.

Keywords Riesz energy · subset selection · computational geometry · parameterized approximation · representative sets · parameterized hardness

1 Introduction

When ten facilities are chosen from thousands of candidates, two nearby sites can contribute more to total interaction than many distant pairs. Riesz energy models this tradeoff in facility location [1, 2] and in the selection of diverse multiobjective solutions [3, 4]. For n sites in a metric space (P, d), exponent $s > 0$ and cardinality k, we minimize

$$E_s(S) = \sum_{\{x,y\} \subseteq S} d(x, y)^{-s}, \qquad \mathrm{OPT}_s(P, k) = \min_{S \subseteq P,\ |S|=k} E_s(S). \tag{1.1}$$

We count unordered pairs. Even two selections tied for max–min p-dispersion can have different energies [5, Proposition 1]. Dropping distant pairs changes the objective, merging close sites risks

Table 1: Theorem 3.1, for fixed d, s and $0 < \varepsilon \leq 1$. Add $O(nk + n \log n)$ preprocessing to the search bound. Constants depend on d, s. Bounds use the exact-operation model; rational inputs and fixed positive even s give polynomial bit cost per operation.

Exponent	Retained candidates	Search over cardinality-k subsets
$0 < s < d$	$O_{d,s}(k^{d/s}/\varepsilon^d)$	$(C_{d,s}k^{d/s-1}/\varepsilon^d)^k$
$s = d$	$O_d(k \log(2k)/\varepsilon^d)$	$(C_d \log(2k)/\varepsilon^d)^k$
$s > d$	$O_{d,s}(k/\varepsilon^d)$	$(C_{d,s}/\varepsilon^d)^k$

a singularity, and the nonincreasing submodular function $-E_s$ lacks the usual monotone greedy guarantee [6].

In the continuous particle model, Hardin and Saff relate minimal Riesz energy to separation and spatial distribution [7, 8]. Prescribing candidate sites changes which moves are feasible. Exact selection is polynomial on lines [9] and ultrametrics [5], but inverse-square decision on rational planar sites is NP-complete [10, Theorem 1.1]. Can we approximate the planar optimum with an input-size exponent independent of k and the accuracy?

There are two separate questions here. Can the dependence on the input size n be confined to preprocessing? If so, how many original sites must still be searched? For the first question one can adapt the proxy lemma of Ceccarello et al. [11, Section 5, Lemma 5] for metric diversity. Its displacement bound, combined with a crude lower bound on optimal Riesz separation, already implies an EPAS; the derivation is given below. Our contribution concerns the second question. An upper bound on the *total* energy is a valid upper bound on one pair, but in the critical planar case this can cost an entire factor of k in the representative count.

Our exchange argument controls only the energy *incident to one selected site*. Choose $2k$ farthest-first centers, and let r be the last insertion distance. The other $k - 1$ sites of an optimum each exclude at most one center at distance strictly below $r/2$. At any of the remaining centers the replacement is feasible, so the selected site's incident energy is at most the average replacement cost. Packing bounds this average by a harmonic sum. Consequently, for $d = s = 2$ every optimum has minimum separation $\Omega(r/\sqrt{\log(2k)})$. This is the right order: in a lattice square with a remote pair, the two remote sites must both be selected even at this small separation (Theorem 3.3). The bound is on the separation of an optimum, not on the spacing of the input.

At this resolution it is safe to keep one original site per occupied cell. The planar bound falls from $O(k^2 \log(2k)/\varepsilon^2)$ for the total-energy packing argument to $O(k \log(2k)/\varepsilon^2)$ for exchange (Proposition 3.6). For fixed accuracy the corresponding search bounds are $2^{O(k \log k)}$ and $2^{O(k \log \log(2k))}$. Neither representative bound depends on n or the aspect ratio. Table 1 gives the other regimes, including a bound linear in k for $s > d$. On a particular input we measure the number M of occupied cells before search: $\binom{M}{k}$ tells us whether direct enumeration fits the budget. A tighter asymptotic bound need not reduce this observed count or make the reduced solve practical.

To see what the generic proxy construction already supplies, write the optimal max–min distance as ρ. Farthest-first covering goes back to Gonzalez [12, Section 2]. For a single input block, the proxy lemma of Ceccarello et al. [11, Section 5, Lemma 5] gives a set of at most $k(8/\varepsilon')^D$ sites in doubling dimension D, with a proxy within $\varepsilon'\rho/2$ of each point of any k-subset. A best-packing selection costs at most $N_k \rho^{-s}$, where $N_k = \binom{k}{2}$, so every pair of a Riesz optimum is at distance at least $\rho/N_k^{1/s}$. Taking $\varepsilon' = \varepsilon/(2qN_k^{1/s})$, $q = \lceil \max(1, s) \rceil$, gives distinct proxies for the optimal sites and the pairwise guarantee of Lemma 3.4: first bound the optimum's separation, then choose the proxy displacement to be small relative to it. Ceccarello et al. do not state a Riesz-energy theorem; this argument uses their single-block proxy property and does not transfer their streaming guarantees.

The use of incident energy also has a continuous antecedent. For $s \geq d$, Hardin and Saff [7, Theorem 2.3, Section 6, equations (59)–(64)] average replacement costs over a region of positive measure. Here the averaging set consists of feasible input sites, even for an arbitrarily irregular finite input. The critical harmonic sum is classical [7, Lemma 3.1]. Continuous subcritical separation results impose regularity hypotheses [13, Theorems 2.3, 2.5]; positive-power diversity schemes [14, Theorem 4] address a different kernel.

Atta's fixed-cardinality grid scheme [1, Sections 6.2–6.4] optimizes a capped objective in $O(n + g^k k^2)$ time for g occupied cells. Its original-energy guarantee assumes minimum input spacing τ and uses cells of diameter below τ; then $g = n$. Our scheme permits arbitrarily close candidates and bounds g independently of n. It is an EPAS jointly in k and $1/\varepsilon$. It does not give a PTAS polynomial in n for fixed accuracy and unrestricted k.

The proof also identifies which part of the geometry is essential. A net replaces the grid in finite metrics of fixed doubling dimension (Theorem 4.2); feasible cell witnesses replace a finite candidate list on compact domains with effective cover and intersection routines, including filled simple polygons (Corollary 4.1). Without a packing restriction the picture changes. At $s = 2$, the two-distance graph encoding of [1, 5] yields W[1]-complete exact decision in k and excludes a deterministic EPAS at accuracy depending on k, unless $\mathrm{FPT} = \mathsf{W}[1]$ (Theorem 5.1). It leaves both fixed-factor approximation in general metrics and exact planar FPT open here.

Appendix F sketches three alternatives at $s = 2$: a Manhattan staircase is isometric to a line after exceptional sites are set aside; projecting a nearly collinear Euclidean core gives a certificate depending on strip width; and a cutoff graph of small treewidth admits exact dynamic programming for the proxy, with an additive bound on the omitted tail. Our experiments in Section 6 use the full energy without imposing any of these three structures.

The paper separates three questions: how much a point of an unknown optimum can move (Section 3), which geometric hypotheses are needed (Sections 4 and 5), and how much search the resulting certificate actually leaves (Section 6). Appendix A collects the notation. For a concrete reading of the proof, Appendix C.1 follows a single optimal site through exchange and rounding. The code and data are available at https://github.com/emmerichmtm/PCGA-Riesz. Appendix G gives the archived inputs and reproduction commands.

2 Model and computational assumptions

The n candidates are distinct. Handle $k \leq 1$ and $k = n$ directly; replace $\varepsilon > 1$ by 1. Dimension d and exponent $s > 0$ are fixed. General-s operation bounds use an exact-real model with arithmetic, comparisons, positive powers and roots, and integer floor as primitives. They do not automatically imply bit bounds for arbitrary real exponents.

For rational Euclidean coordinates and fixed positive even s, all weights are rational. The algorithm below uses rational arithmetic, comparisons, binary lengths, and integer division, with polynomial bit cost in total input length L (including rational accuracy). It avoids constructing all $\binom{n}{2}$ input weights. A deterministic EPAS returns a feasible $(1 + \varepsilon)$ approximation in $f(k, 1/\varepsilon)L^c$ bit time for an absolute constant c. A bound $n^{g(k)}$ establishes only XP. Our representative set contains at least one near-optimal selection for the requested k; it need not preserve every subset or bound coordinate encoding length. It is not a bounded-bit kernel.

For exact decision, the input also includes a rational threshold. We use parameterized reductions and the classes $\mathrm{FPT} \subseteq \mathsf{W}[1] \subseteq \mathsf{W}[\mathsf{P}]$ [15]. W[1]-hardness provides conditional evidence against exact FPT; completeness additionally requires membership. Approximation hardness will require a separate gap argument.

3 Geometric compression by oversampling and exchange

Put $H_m^{(p)} = \sum_{j=1}^{m} j^{-p}$. Throughout this section $d \geq 1$ and $s > 0$ are fixed, $2 \leq k < n$, and $0 < \varepsilon \leq 1$.

Theorem 3.1 (Fixed-dimensional parameterized approximation)**.** *For $P \subset \mathbb{R}^d$, a deterministic algorithm retains $Q \subseteq P$ with*

$$k \leq M := |Q| \leq n, \qquad M = O_{d,s}\left(\frac{k\big(H_{2k}^{(s/d)}\big)^{d/s}}{\varepsilon^d}\right), \tag{3.1}$$

and computes its exact cardinality-k optimum V using

$$O(nk + n\log n) + \left(\frac{C_{d,s}\big(H_{2k}^{(s/d)}\big)^{d/s}}{\varepsilon^d}\right)^k \tag{3.2}$$

operations in the model of Section 2. The returned original candidates satisfy

$$V/(1+\varepsilon) \leq \mathrm{OPT}_s(P,k) \leq V. \tag{3.3}$$

No input separation, bounded aspect ratio, or partition is required. For rational coordinates and fixed positive even s, each operation has polynomial bit cost.

For $d = s = 2$, Proposition 3.6 compares this bound with the two total-energy estimates on the same footing. Theorem 3.3 identifies the step that cannot be improved in general: separation at the cover scale.

A representative set must have two properties. It must contain a distinct representative for each point of some near-optimal selection, and it must be small enough to search. The first property is delicate because the kernel diverges at a collision. We establish it by finding a length scale for an *optimum*, without imposing a minimum spacing on the input.

3.1 A finite set of feasible replacements

If $n \leq 2k$, retain P and enumerate directly; this fits the theorem because $\binom{n}{k} \leq 4^k$. Hence suppose $n > 2k$. Choose $2k$ centers $C \subset P$ by farthest-first traversal [12], starting arbitrarily, and let $r > 0$ be the last insertion distance. Insertion radii are nonincreasing, so C is r-separated. Before the final insertion all candidates are within r of the previous centers; thus the radius-r balls around C cover P. Maintaining nearest-center distances takes $O(nk)$ operations. Squared rational distances suffice for all comparisons.

Lemma 3.2 (Replacement potential and separation)**.** *Let a feasible set P contain the $2k$ mutually r-separated centers, and let S be a cardinality-k subset admitting no improving one-point replacement from P. For every $x \in S$,*

$$\sum_{y \in S\setminus\{x\}} \|x-y\|^{-s} \leq \frac{4^s H_{2k}^{(s/d)}}{r^s}. \tag{3.4}$$

In particular, every optimum is separated by

$$\delta = \frac{r}{4\big(H_{2k}^{(s/d)}\big)^{1/s}}. \tag{3.5}$$

Proof. Fix x, put $A = S \setminus \{x\}$, and let

$$T_x = \{c \in C : \|c-y\| \geq r/2 \text{ for all } y \in A\}.$$

Each y excludes at most one center at distance *strictly less than* $r/2$, since two such centers would be less than r apart. Therefore $|T_x| \geq k+1$. Every $c \in T_x$ lies outside A and is a feasible replacement; $c = x$ is harmless. With $\Psi_A(z) = \sum_{y\in A} \|z-y\|^{-s}$, replacement optimality gives $\Psi_A(x) \leq \Psi_A(c)$.

For fixed $y \in A$, sort the free-center distances as $r/2 \leq \rho_1 \leq \cdots \leq \rho_{|T_x|}$. The disjoint open radius-$r/2$ balls about the first j centers lie in the radius-$(\rho_j + r/2)$ ball about y. Volume comparison yields

$$j \leq (1+2\rho_j/r)^d \leq (4\rho_j/r)^d, \qquad \rho_j^{-s} \leq 4^s r^{-s} j^{-s/d}.$$

The point y need not be a center. Averaging over the feasible replacements gives

$$\begin{aligned}\Psi_A(x) &\leq \frac{1}{|T_x|}\sum_{c\in T_x}\sum_{y\in A}\|c-y\|^{-s} \\ &\leq \frac{k-1}{|T_x|}\,4^s r^{-s} H_{2k}^{(s/d)} \leq 4^s r^{-s} H_{2k}^{(s/d)}.\end{aligned}$$

Every individual pair term is at most this potential, proving (3.5). □

The averaging removes the factor k lost by bounding one pair with the entire incumbent energy. Only one-swap optimality is used: the algorithm need not run local search. The lemma bounds separation; it does not bound the energy ratio of a local optimum or the time needed to find one. Improving one-point replacements also underlie the practical heuristic of [3, Algorithm 3]; that algorithm has no approximation bound.

Theorem 3.3 (Sharpness of planar separation)**.** *For every sufficiently large k there is a finite rational $P_k \subset \mathbb{R}^2$ such that, for any start and tie-breaking of farthest-first traversal, every one-swap local optimum for E_2, and hence every global optimum, has minimum distance $\Theta(r_{2k}/\sqrt{\log k})$. Here r_{2k} is the last insertion distance among $2k$ centers; the constants are absolute.*

The construction consists of a square of unit lattice sites and two sites far from that square but close to each other. If one remote site is omitted, almost all selected sites lie in the square. Their interactions across successive spatial scales force some site to have incident energy at least $c\log k$. Moving it to the unoccupied remote site costs strictly less, even if the other remote site is already selected. Thus both members of the close pair must occur. Appendix B.1 gives the construction and the counting argument. This shows sharpness of the separation lemma; it gives no lower bound on representative-set size or on the running time of a different approximation scheme.

3.2 Injective rounding and exact search

Once a separation bound is known, no knowledge of the optimal sites themselves is needed. We choose the grid first and keep any one feasible site per occupied cell. The next lemma shows why this cannot merge two sites of an optimum.

Lemma 3.4 (Rounding an optimum)**.** *Suppose every pair of sites in an optimal selection is at distance at least a known $\sigma > 0$. Let $q \geq \max\{1, s\}$ be a fixed integer, choose $\sigma/2 < \widehat{\sigma} \leq \sigma$, and grid $\mathbb{R}^d$ into half-open cubes of side $h = \varepsilon\widehat{\sigma}/(4qd)$. Retain one feasible point from each cell intersecting the feasible set. The resulting Q contains a $(1+\varepsilon)$-approximate k-subset. A cover by c sets of diameter at most a gives $|Q| \leq c(2+a/h)^d$.*

Proof. Cell diameter is $\sqrt{d}\,h < \sigma$, so snapping an optimum to retained candidates is injective. A selected distance $D \geq \sigma$ decreases by at most $2\sqrt{d}\,h \leq \varepsilon D/(2q)$. For $s \geq 1$, Bernoulli's inequality gives $(1-\varepsilon/(2q))^s \geq 1-\varepsilon/2$; for $s < 1$, use $(1-z)^s \geq 1-z$. Since $1-\varepsilon/2 \geq (1+\varepsilon)^{-1}$ for $0 \leq \varepsilon \leq 1$, every pair energy increases by at most $1+\varepsilon$. Sum over pairs. Each diameter-a set has coordinate ranges at most a and meets at most $(2+a/h)^d$ cells. □

Proof of Theorem 3.1. Apply Lemma 3.4 with $\sigma = \delta$ from Lemma 3.2. The $2k$ covering balls have diameter $2r$, so

$$M \le 2k(2 + 2r/h)^d = O_{d,s}\Big(k\big(H_{2k}^{(s/d)}\big)^{d/s}/\varepsilon^d\Big).$$

In the exact real model we may take $\widehat{\sigma} = \sigma$; rational implementation is described below. Sort cell keys in $O(n \log n)$ operations. Evaluate each of the $\binom{M}{k}$ subsets in $O(k^2)$ operations and retain the best. The bound $\binom{M}{k} \le (eM/k)^k$, with $k^2 \le 4^k$, proves (3.2). Optional precomputation of representative pair weights is absorbed by this bound. Since $Q \subseteq P$ and the snapped optimum is feasible in Q, equation (3.3) follows. Any additional feasible solution of energy U can improve the upper endpoint to $\min\{U, V\}$.

For rational coordinates and positive even s, write $m = 2k$, $b = \lceil \log_2 m \rceil$, and use the rational upper bound

$$\overline{H} = \begin{cases} \dfrac{d}{d-s}\, 2^{\lceil b(d-s)/d \rceil}, & s < d, \\ 1 + b, & s = d, \\ \dfrac{s}{s-d}, & s > d. \end{cases} \tag{3.6}$$

Integral bounds give $H_m^{(s/d)} \le \overline{H} = O_{d,s}(H_m^{(s/d)})$. Thus $B = 4^s \overline{H}/(r^2)^{s/2}$ is rational and $\sigma = B^{-1/s}$ is a valid separation scale with the same asymptotic bounds. Binary lengths and a constant number of rational tests find a dyadic $\widehat{\sigma}$ satisfying $B\widehat{\sigma}^s \le 1 < B(2\widehat{\sigma})^s$. No root is computed. Grid indices use integer division. All coordinates, keys and sums of at most $\binom{k}{2}$ rational pair weights have polynomial bit length. Appendix B.2 gives the arithmetic details. □

Corollary 3.5 (Planar inverse-square scheme)**.** *For rational planar input and $s = 2$, the scheme retains $O(k \log(2k)/\varepsilon^2)$ candidates and uses*

$$O(nk + n \log n) + (C \log(2k)/\varepsilon^2)^k$$

operations of polynomial bit cost, for an absolute constant C.

The three regimes in Table 1 follow from $H_m^{(p)} = O_p(m^{1-p})$ for $p < 1$, $O(\log m)$ for $p = 1$, and $O_p(1)$ for $p > 1$. At fixed accuracy, planar $s = 2$ has search bound $2^{O(k \log\log(2k))}$; $s > d$ has $2^{O_{d,s}(k)}$. These remain exponential searches, not unrestricted PTASs. The rounding uses a single grid scale; adaptive representatives might still improve its size.

3.3 Comparing planar separation certificates

Let C_k be the first k farthest-first centers, let r_k be their last insertion radius, and put $U = E_2(C_k)$ and $N_k = \binom{k}{2}$. For any optimum, a single pair costs at most the optimum's total energy, which is at most U; thus $B \ge U$ certifies separation $B^{-1/2}$. An incident-energy bound can certify separation without bounding U. Table 2 lists explicit choices; r_{2k} is the radius used by Lemma 3.2.

Proposition 3.6 (Planar comparison of certificates)**.** *The pair-count and incumbent-packing analyses give $O(k^3/\varepsilon^2)$ and $O(k^2 \log(2k)/\varepsilon^2)$ representatives, respectively. The exchange analysis gives $O(k \log(2k)/\varepsilon^2)$. All preserve a $(1 + \varepsilon)$ approximation to the original objective.*

Proof. The r_k-separated incumbent satisfies $U \le N_k/r_k^2$. For a center's jth nearest companion at distance d_j, packing gives $j + 1 \le (1 + 2d_j/r_k)^2$, hence $j \le 8(d_j/r_k)^2$ because $d_j \ge r_k$. Summing over companions and dividing by two yields $U \le 4kH_{k-1}/r_k^2$. The k covering disks and Lemma 3.4 give $M = O(kr_k^2 B/\varepsilon^2)$ in either case. For exchange, the $2k$ covering disks have radius r_{2k}, and Lemma 3.2 bounds each selected site's incident energy by $16H_{2k}/r_{2k}^2$. That radius again cancels from the count, yielding the third row. Enumeration uses $\binom{M}{k} \le (eM/k)^k$. □

Table 2: Explicit planar certificates and bounds. The first two rows bound the total energy of the farthest-first selection. The third bounds the energy incident to one site of an optimum. $H_j = \sum_{i=1}^{j} 1/i$. B is an upper bound on every optimum pair contribution.

Certificate	B	Representative upper bound
Pair count	N_k/r_k^2	$O(k^3/\varepsilon^2)$
Incumbent packing	$\min\{N_k, 4kH_{k-1}\}/r_k^2$	$O(k^2\log(2k)/\varepsilon^2)$
Finite exchange	$16H_{2k}/r_{2k}^2$	$O(k\log(2k)/\varepsilon^2)$

A certified hybrid. Let $B_X = 16H_{2k}/r_{2k}^2$, and let $\overline{U} \geq U$ be any evaluated incumbent upper bound. Use $B_H = \min\{\overline{U}, B_X\}$ in the same dyadic rounding rule. Both bounds are valid, so the guarantee is unchanged. Since $B_H \leq B_X$, the new width is no smaller than the exchange width; the $2k$ covering disks prove the same asymptotic bound. With one common physical grid origin, dyadic widths are nested, so the hybrid retains at most the smaller of the two representative counts. This concerns counts and certificates, not dominance of returned energies. Computing U costs $O(k^2)$ operations and does not change the theorem's bound. Appendix B.3 proves the nesting and gives a bounded-denominator evaluation of $\overline{U}$.

Deciding whether to enumerate. For fixed d, s, the EPAS is parameterized by k and $1/\varepsilon$. For a given input, sorting the cell keys gives the actual number M of representatives before subset search. For a budget of T complete subset evaluations, enumerate if $\binom{M}{k} \leq T$. At $T = 10^7$ the largest admissible M is 67 for $k = 5$ and 27 for $k = 10$. Each evaluation sums $\binom{k}{2}$ pair weights. A larger count may still admit an exact solution by pruning, as in the Swiss example. Completing the exact reduced solve gives the certificate in (3.3).

The certified cell side h sets the scale at which clustering matters. If each of c clusters lies in a distinct single cell, then $M = c$. When c is close to k, this leaves few extra representatives. A cluster of diameter below h can still meet up to 2^d cells at grid boundaries. For $n > 2k$ the exchange grid occupies at least $2k$ cells because its cell diameter is smaller than the separation r of its $2k$ centers. With fewer than $2k$ tight groups, traversal must enter some group twice and can produce a very fine grid. The hybrid in Section 3.3 can give $M = k$, leaving a single reduced choice. In a uniformly sampled region with G cells of positive sampling probability, $M \leq \min\{n, G\}$. For a fixed grid, dense sampling fills an increasing fraction of those cells. Appendix B.4 quantifies both effects.

The constants explain a limitation of comparing these grids by completed solves. On the exchange branch ($n > 2k$), $r_{2k} \leq r_k$. The implemented incumbent bound is capped by N_k/r_k^2, whereas $B_X = 16H_{2k}/r_{2k}^2$. For $2 \leq k \leq 11$ one has $N_k < 16H_{2k}$, so the incumbent grid is always at least as coarse as exchange at a common origin; the hybrid then uses the incumbent grid. At the other end, $M_X \geq 2k$ and $\binom{26}{13} = 10{,}400{,}600 > 10^7$, so at $k \geq 13$ the exchange grid alone always exceeds this enumeration budget. At $k = 12$ an additional planar estimate proves the exact incumbent is still tighter than exchange (Appendix B.3). An observed reduction in M at larger k can still benefit a search that prunes; it does not by itself establish that such a search finishes.

4 Beyond finite Euclidean inputs

The proof used feasible replacements, packing, and a way to find one representative near each optimum point. These ingredients can be separated. On a filled polygon the candidates are no longer listed, so we must construct feasible points in cells. In a finite metric the sites are listed, but there is no coordinate grid; a separated net takes its place.

Corollary 4.1 (Compact feasible domains)**.** *Let $D \subset \mathbb{R}^d$ be compact. Suppose one can compute $2k$ feasible centers at mutual distance at least $r > 0$ whose closed radius-r balls cover D, and can test a half-open grid cell for intersection with D and return a feasible witness when it intersects. In the exact-operation model, retaining one witness per intersecting cell gives a $(1+\varepsilon)$-approximation with the representative bound (3.1) (omitting $M \le n$). The cost is $T_{\mathrm{cert}} + T_{\mathrm{grid}} + O(k^2\binom{M}{k})$, where T_{grid} includes scale computation, cell enumeration, and witness queries.*

Proof. On D^k extend the energy to $+\infty$ at collisions. It is lower semicontinuous and some feasible tuple has finite energy, so its minimum is attained at distinct points. The given centers are feasible replacements in Lemma 3.2, which uses only their separation. Lemma 3.4 applies to the witness in each optimal point's cell. Enumerate the grid keys in the axis-aligned bounding boxes of the $2k$ covering balls, remove repeated keys, and query the remaining cells. Each box yields at most $(2 + 2r/h)^d$ keys. The ratio r/h is $O_{d,s}((H_{2k}^{(s/d)})^{1/s}/\varepsilon)$, so the absolute scale cancels from the number of cell queries. For $s = d = 2$ the algorithm queries $O(k\log(2k)/\varepsilon^2)$ cells, followed by at most $(C\log(2k)/\varepsilon^2)^k$ operations for subset search. □

In a closed simple polygon with m vertices, a continuous farthest-first step with j sites maximizes the empty-circle radius subject to its center lying in the polygon. Chew and Drysdale's Voronoi algorithm uses $O(mj + j\log(2j))$ geometric operations for this step [16]. A triangulation supplies cell witnesses by clipping triangles to the closed cell and testing the excluded upper faces. Appendix B.6 details the boundary tests and coordinate bit costs. For polyhedral domains in higher dimensions the corollary applies with the costs of the supplied certificate and witness routines.

Theorem 4.2 (Finite metrics of bounded doubling dimension)**.** *Let P be an n-point metric in which every closed radius-R ball is covered by at most 2^λ closed radius-$R/2$ balls with centers in P. For fixed $\lambda \ge 1$ and $s > 0$, an exact distance oracle and the stated power and comparison primitives suffice for a deterministic EPAS. It retains*

$$M = O_{\lambda,s}\Big(k(H_{2k}^{(s/\lambda)})^{\lambda/s}/\varepsilon^\lambda\Big)$$

input sites. Center and net construction take $O(nk + nM)$ distance queries. Enumeration takes $O(k^2\binom{M}{k})$ distance and arithmetic operations.

Proof. For $n \le 2k$ enumerate the input directly. Otherwise use its $2k$ farthest-first centers. For an a-separated set in a closed radius-R ball, if $R < a/2$ there is at most one point. If $R \ge a/2$, iterate the doubling cover $\ell = \lfloor \log_2(2R/a) \rfloor + 1$ times. Each resulting ball has diameter strictly below a, including when $2R/a$ is a power of two. The number of points is at most $2^{\lambda\ell} \le (4R/a)^\lambda$.

Fix an optimal site x and its $k-1$ companions. Their open radius-$r/2$ balls exclude at most $k-1$ centers. For each companion the jth nearest remaining center has distance $\rho_j \ge r/2$, so $j \le (4\rho_j/r)^\lambda$. Averaging the exchange inequalities over the at least $k+1$ remaining centers, as in Lemma 3.2, bounds the incident potential by $4^s r^{-s} H_{2k}^{(s/\lambda)}$. Thus every optimal pair has distance at least $\delta = r/[4(H_{2k}^{(s/\lambda)})^{1/s}]$.

Put $\eta = \varepsilon\delta/(4q)$ with integer $q \ge \max(1, s)$. Scan P, retaining a site when all earlier representatives are farther than η. Every site has a representative within η. Rounding an optimal pair at distance $b \ge \delta$ leaves distance at least $b - 2\eta \ge (1 - \varepsilon/(2q))b > 0$. This proves injectivity and, by the scalar inequality in Lemma 3.4, the energy guarantee. The η-separated representatives in each covering ball number at most $(4r/\eta)^\lambda$. Summing over the $2k$ balls proves the bound. A scan compares each input site with at most M representatives. □

For $s < \lambda$ the count is $O_{\lambda,s}(k^{\lambda/s}/\varepsilon^\lambda)$. At $s = \lambda$ it is $O_\lambda(k\log(2k)/\varepsilon^\lambda)$. For $s > \lambda$ it is $O_{\lambda,s}(k/\varepsilon^\lambda)$. Here λ bounds the doubling dimension of the metric supplied to the oracle. Section 3

instead uses ambient Euclidean volume. The net is the familiar diversity proxy construction [11, Section 5]. The energy-specific step is the bound on the potential of an optimal site, which determines how far a proxy may move while preserving the entire inverse-power sum.

Consistency with best-packing. For fixed $k \geq 2$, write $\Delta(S)$ for the minimum pair distance and $N = \binom{k}{2}$. The bounds $\Delta(S)^{-s} \leq E_s(S) \leq N\Delta(S)^{-s}$ give $E_s(S)^{1/s} \to 1/\Delta(S)$. Morris and Mitchell use an inverse-distance criterion for maximin experimental designs [17]. The limit recovers best-packing [5, Proposition 1]. For compact sets see also [18, Proposition 2]. At finite s, Lemma 3.2 supplies the scale that controls all pair terms simultaneously. Our complexity bounds fix s.

5 What fails in unrestricted metrics

The doubling assumption controlled how many separated replacements fit near any one companion. Without a packing restriction, even a metric with just two nonzero distances can encode an arbitrary graph. This gives both an exact classification and a separate obstruction to approximation.

Theorem 5.1 (Exact classification and approximation barrier)**.** *At $s = 2$, exact Riesz selection on metrics with nonzero distances in $\{1, 2\}$ is W[1]-complete parameterized by k. Unless $\mathrm{FPT} = \mathrm{W}[1]$, even this restriction has no deterministic $(1 + \varepsilon)$ approximation in $f(k, 1/\varepsilon)L^{O(1)}$ time.*

Proof. Use Atta's two-distance reduction [1, Section 5], equivalently the complementary-clique construction in [5, Theorem 1]. For a graph G, assign distance 1 on edges and 2 on nonedges. Every positive distance is at most the sum of two positive distances, so this is a metric. For $N_k = \binom{k}{2}$ and a k-set S,

$$E_2(S) = N_k/4 + (3/4)e_G(S). \tag{5.1}$$

The objective is an affine transformation of Sparsest k-Subgraph, whose general-graph W[1]-hardness is documented in [19]. Thus threshold $B = N_k/4$ is attainable exactly when G has a k-independent set. The reduction preserves k, proving W[1]-hardness.

For membership at any rational threshold, enumerate the 2^{N_k} labeled distance patterns on $[k]$ and discard those exceeding the threshold. For each remaining pattern, form a k-partite graph with vertices (i, u), $i \in [k]$, $u \in P$. Join (i, u) and (j, v) precisely when $i \neq j$, $u \neq v$, and their input distance matches the pattern on ij. A k-clique realizes the pattern with distinct candidates. The disjoint union of these graphs reduces the problem to k-Clique in $f(k)L^{O(1)}$ time, with unchanged parameter. For $k \geq 2$ a clique lies in one component; smaller cases are direct. This proves W[1] membership.

For the approximation barrier, assume $k \geq 2$ and set $\varepsilon = 1/N_k$. A yes-instance has optimum B, so a $(1 + \varepsilon)$ algorithm returns energy at most $B + 1/4$. A no-instance has every feasible energy at least $B + 3/4$. Comparing the returned energy with $B + 1/2$ decides Independent Set. The no/yes optimum ratio is at least $1 + 3/N_k$; taking $1/\varepsilon = N_k$ keeps the runtime parameter a function of k alone. Thus the assumed scheme would decide Independent Set in FPT time. □

For arbitrary rational weights, guessing k indices uses $O(k \log n)$ nondeterministic bits and permits polynomial-time exact verification. The bounded-nondeterminism characterization gives W[P] membership [15]; it does not give W[1] membership when weight precision is unrestricted. Table 3 separates these statements.

The guaranteed relative gap $3/N_k$ tends to zero. On the hard subclass every feasible k-subset is a 4-approximation because its energy lies between $N_k/4$ and N_k. The reduction therefore does not exclude a fixed-factor approximation, or a scheme whose input-size exponent depends on accuracy. Nor does it establish W[1] membership for exact selection in all rational metrics. The hard family

Table 3: Subset-size complexity at $s = 2$. Approximation uses the joint parameter $k + 1/\varepsilon$; exact decision uses k. Rational encodings are assumed.

Setting	Exact decision	Approximation
Metrics with distances $1, 2$	W[1]-complete	No deterministic EPAS unless FPT = W[1]; trivial factor 4
General rational metrics	W[1]-hard; in W[P]	Same obstruction
Bounded doubling (rational distances)	In W[P]	EPAS for fixed λ, Theorem 4.2
Rational Euclidean plane	In W[P]; exact FPT status not settled here	EPAS, Corollary 3.5

has no uniform doubling bound: for an edgeless graph the radius-two ball contains all n sites and each radius-one ball is a singleton. The reduction does not establish planar W[1]-hardness. Conversely, the planar EPAS does not settle exact FPT in the plane: isolating an exact rational threshold can require accuracy depending on input bit length.

6 Certificates, search cost and experiments

We distinguish three outcomes: a smaller representative set, a completed exact solve on that set, and a comparison with an independently computed full-input optimum. Counting occupied cells establishes the first; it does not establish either of the other two. We compare the certificates of Section 3.3 on the same inputs and measure what refinement adds after a completed reduced solve.

6.1 Certified post-processing

Proposition 6.1 (EPAS-Refine)**.** *Starting at the exact reduced solution of energy V, visit the selected cells in a fixed order. In each, choose an original candidate of minimum potential against the other current sites, retaining the current site on a tie. After t complete sweeps,*

$$V/(1+\varepsilon) \le \mathrm{OPT} \le U_t \le U_{t-1} \le \cdots \le U_0 = V \le (1+\varepsilon)\,\mathrm{OPT}\,.$$

Grouped cell membership gives $O(tnk)$ additional arithmetic operations.

Proof. The k selected cells are distinct. Replacing one site changes precisely its $k-1$ incident terms, whose sum is minimized over a set containing its current position. Feasibility and the upper bound follow by induction. Within a sweep the selected cells are disjoint, so at most n candidates each require $k-1$ pair evaluations. The lower bound follows from (3.3). □

6.2 Search costs and solution quality

We measure occupancy, exact reduced search and within-cell refinement at $d = s = 2$ and $\varepsilon \in \{1/2, 1\}$ (energy factors 1.5 and 2). The Swiss sites use whole-metre national coordinates and the synthetic benchmarks use dyadic coordinates. The implementation counts integer cell keys before allocating reduced pair weights and uses rational branch and bound. Swiss refinement compares exact fractions. On the synthetic benchmarks, outward-rounded intervals filter candidates before exact comparison. Appendix B.5 details the numerical safeguards; Appendix E describes 157 validation checks.

Dispersion across Swiss settlements. Figure 1 shows $k = 10$ facilities chosen from 5,403 synthetic sites. The point clouds follow 26 cantonal seats and ten rural settlements. A single candidate near the eastern border represents Müstair. At $\varepsilon = 1$, the exchange grid retains 3,365

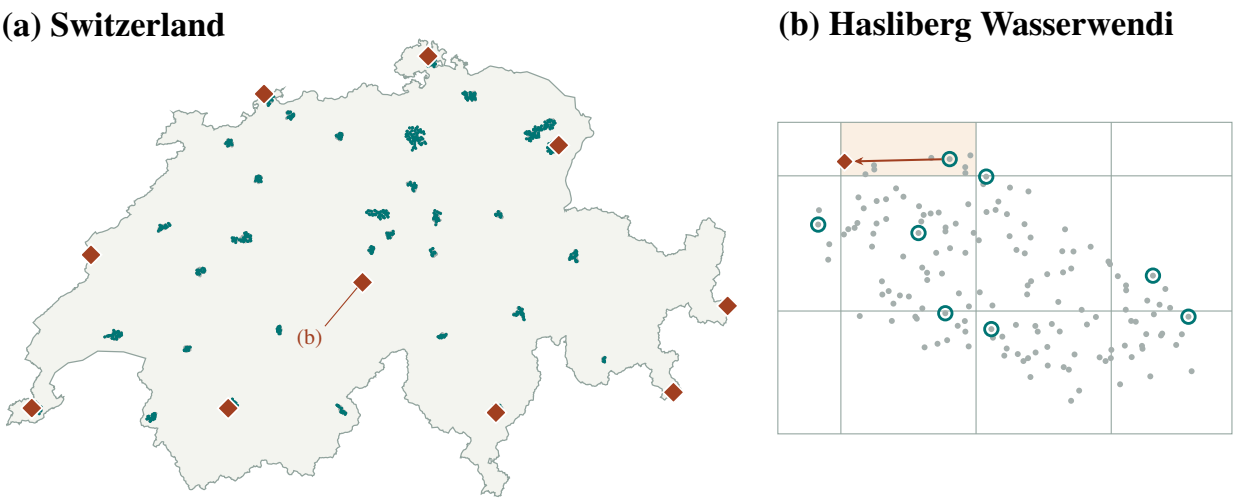


Figure 1: Ten facilities selected from 5,403 synthetic Swiss sites ($s = 2$, $\varepsilon = 1$). (a) Grey points are candidates, teal points are the 706 hybrid grid representatives, and rust diamonds are the refined facilities, verified optimal on the full input. (b) At Hasliberg Wasserwendi, 150 candidates occupy eight cells of side 1,024 m; rings mark their representatives. The arrow shows the replacement within the shaded cell. Settlement geometry and outline: swisstopo [20, 21].

sites (cell side 256 m). The hybrid instead retains one representative per occupied 1,024 m cell, leaving 706 sites (86.93% removed). Even after compression there are $\binom{706}{10} \approx 7.95 \times 10^{21}$ possible selections. A diameter bound ruled out taking two sites from one locality. Lower bounds on interactions between localities left only 69 sets of ten localities for further exact search.

At this $k = 10$ the hybrid grid is necessarily the incumbent grid. Its improvement from 3,365 to 706 representatives is due to the evaluated incumbent; the completed solve owes its feasibility to locality pruning. The hybrid representative optimum is 0.32553% above the full optimum. Eight within-cell replacements recover the latter exactly, at energy $2.81234821 \times 10^{-9}\,\mathrm{m}^{-2}$. One move selects a site in Hasliberg Wasserwendi, 5.87 km from the national area centroid, without a central-coverage constraint. The EPAS guarantees a factor of two here; exact recovery is an observation proved by a separate full-input search. Appendix D gives the construction and exact audit, with commands for reproducing both.

Could the small Swiss gap be certified without solving the full instance? One route is to compute a lower bound on the full optimum. A cutoff drops the cost of distant pairs; a stepwise lower envelope instead rounds distances up to bin endpoints. With a diameter upper bound D as the last endpoint, every selection contributes the same baseline $\binom{k}{2} D^{-s}$; only pairs with a positive residual need edges. Given a width-w decomposition of this graph, the dynamic program in Appendix F computes the proxy optimum B, including the baseline, in $2^{O(w)} n(k+1)^2$ operations after preprocessing. For reduced optimum V and any feasible energy U, the bound $L_* = \max\{B, V/(1+\varepsilon)\}$ certifies $L_* \le \mathrm{OPT} \le U$. Thus $U \le 1.01 L_*$ would certify one per cent accuracy on that instance. Choosing bins that give a useful bound at small width remains an open problem.

Density and search cost. With $n = 10^6$, $k = 5$ and $\varepsilon = 1/2$, the exchange grid had side $h = 1/256$ on all six inputs. Each input with ten tight lattice clusters gave $M = 10$ and $\binom{10}{5} = 252$ subsets. Median preprocessing and exact search took 0.377 s and 0.00326 s. Uniform inputs in $[0, 4)^2$ occupied 644,670 to 645,259 cells, leaving about 9.3×10^{26} subsets. With 10^5 sites in just five tight groups, the hybrid reached $M = k = 5$ on all three seeds when the groups lay inside cells.

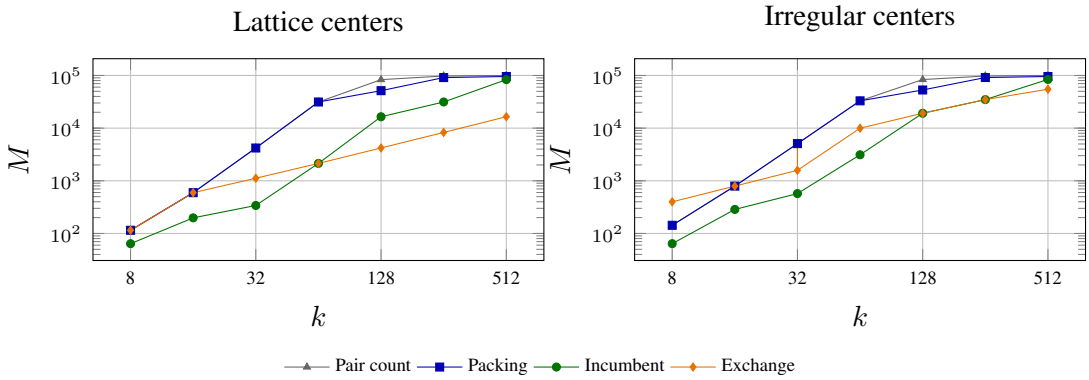


Figure 2: Representatives retained at $\varepsilon = 1/2$ on $n = 10^5$ sites in $2k$ groups of half-width $1/256$. Curves show medians over three seeds; vertical segments span the full range. All four certificates use the same points and grid origin. The hybrid count is the smaller of incumbent and exchange. No subset search was attempted on these inputs.

The exchange grid retained over 89,000 sites on the same inputs. Table 6 gives both grid widths and shows what happens when their common origin moves.

Paired comparison. We compare pair count (PC), packing (PK), an evaluated incumbent (I), exchange (X), and the hybrid (H) at $\varepsilon = 1/2$. The 114 paired configurations give 570 method runs on 99 distinct input arrays. All five use identical inputs, traversal, physical grid origin, first-index representatives and search limits. PC and PK use Table 2. I rounds the incumbent energy upwards with error below $\binom{k}{2}2^{-48}$. H uses the tighter I/X bound. We report medians and ranges over seeds $17, 29, 43$.

The $n = 2000$, $k = 5$ inputs vary group shape, location, population, width and grid origin, and include uniform controls. Nine $n = 24$, $k = 4$ inputs also permit exhaustive checks of the original optimum. Across the 72 search-eligible cases per method, we attempt search when $M \le 160$ and $\binom{M}{k} \le 10^7$, with a five-second cap. PC/PK/I/X/H completed 60/60/60/48/60 searches, respectively. The 72 remaining method runs exceeded a size limit.

At $k = 5$, irregular groups of half-width $1/1024$ retained median counts of 18, 18, 16, 40, 16 for PC, PK, I, X and H, respectively. The evaluated incumbent gave a coarser grid than exchange here. The hybrid preserves that practical benefit and the exchange theorem's worst-case bound. For $n = 10^5$, the occupancy study varies $k = 8, 16, \ldots, 512$ together with the number of groups, $2k$. At $k = 512$ the lattice medians in Figure 2 are $M_I = 83{,}390$ and $M_X = 16{,}458$; the irregular-center medians are $84{,}091$ and $54{,}751$. Exchange gives the coarser grid for both families at this endpoint, but the reduced sets still preclude enumeration. The split by task matters. Among the 72 cases with a search protocol, X retained fewer/equal/more sites than I in $0/13/59$ cases. The 60 searches completed by both H and I had identical ordered representative lists and exact reduced energies. Among the 42 occupancy-only cases, the split was $12/9/21$; all twelve strict improvements occurred at $k = 128, 256$ or 512. None shows a completed solve made feasible solely by exchange.

To separate the effect of k from a change in the data, we also varied k on six frozen point arrays (Table 8). On the 24-group arrays, I and H each completed 13 of 24 searches and X completed six. On the same three 256-group arrays at $k = 128$, X consistently used fewer representatives: the median fell from 1,004 to 933. When we removed both size gates and actually launched all nine I/X/H searches, none finished in five seconds. At $k = 192$ the median ordering reversed.

Exchange improves the worst-case bound and can shrink the observed grid on fixed points; these measurements do not demonstrate an exchange-only gain in completed searches.

How many refinement sweeps help? We compared sweep budgets $0, 1, 2, 3, 5, 10$ on the 84 completed searches of the 156-run scale study. All budgets follow the same deterministic trace from the exact exchange-grid optimum; after a sweep without improvement the state and time are carried forward. In 78 runs the first sweep attained the final recorded energy; in the other 6, the second sweep did. The median decrease from the reduced energy was 0.05231% (maximum 0.35347%). Median extra time was 2.94 ms for one sweep and 5.48 ms at the ten-sweep cap. Refinement attained the full optimum on 7 of 9 small audited cases. The 3 timeouts and 69 budget exclusions supplied no refinement observations. Against a two-sweep global coordinate heuristic the refined energy was lower/equal/higher in 26/47/11 completed cases. Appendix E.1 gives the sweep table and separate fixed-seed checks.

7 Discussion and conclusion

For fixed accuracy, our EPAS reduces Riesz-energy selection to a search whose size depends on k, d, s alone, however many candidates are offered. Reading and grouping a finite input still costs $O(nk + n \log n)$ operations. The independence from n has a geometric consequence: on a filled simple polygon, a point from each cell that meets the polygon suffices for the same finite search, even though there are infinitely many possible locations. A net gives the analogous result for finite metrics of bounded doubling dimension.

Generic proxies already give an EPAS. The exchange argument sharpens the worst-case estimate for the distance an optimal site can move under rounding. Averaging the incident energy over feasible replacements improves the planar inverse-square representative bound from $O(k^2 \log(2k)/\varepsilon^2)$ for the total-energy packing estimate to $O(k \log(2k)/\varepsilon^2)$. The lattice-and-pair construction shows that the logarithm in the separation estimate is unavoidable at this cover scale. It leaves room for adaptive rounding that exploits more than minimum separation. Without geometric packing, even two distances suffice for W[1]-complete exact decision and rule out a deterministic EPAS under $\mathrm{FPT} \neq \mathsf{W}[1]$. The obstruction uses accuracy tending to zero; it settles neither fixed-factor approximation in general metrics nor exact FPT in the Euclidean plane.

In practice one can compute both certificates, keep the coarser grid, then inspect M and $\binom{M}{k}$ before committing to a search. A million clustered sites left just 252 five-site subsets; a million uniform sites left about 9.3×10^{26}. Exchange sometimes reduced M on the same data at larger k, but none of those gains produced a newly completed solve in our experiments. The Swiss instance illustrates a different opportunity: the incumbent grid and locality pruning made a ten-site solve possible, and eight within-cell moves attained the independently verified full optimum. The general certificate allowed a factor of two. A lower bound computed from the occupied cells or from a sparse interaction proxy could certify such accuracy on an individual instance without solving the full problem.

Use of AI tools

OpenAI ChatGPT/Codex assisted with proofreading, implementing and testing the algorithms, running computational experiments, and preparing the figures. The authors take responsibility for the technical claims, algorithmic ideas, references and interpretations.

A Table of symbols

The table collects the notation used throughout the main argument. Local variables in individual proofs and in the exact-search implementation are defined where used. For the proofs and code, Appendix B gives the arithmetic, Appendices C–E give the worked examples and experiments, and Appendix G gives a single command to replay them. Appendix F treats structural alternatives.

Table 4: Notation for selection, geometric compression and certification.

Symbol	Meaning
Problem and computational model	
$P,\ n$	Feasible input sites and their number; in the continuous extension the feasible set is a compact domain D.
$S,\ k,\ S^\star$	A selection, its prescribed cardinality and an optimal selection.
$d,\ s$	Euclidean dimension and positive Riesz exponent. In metric notation $d(x, y)$ denotes distance.
$E_s(S),\ \mathrm{OPT}_s(P, k)$	Energy summed over unordered pairs and the minimum energy of a feasible k-set.
$N_k,\ \Delta(S)$	Number $\binom{k}{2}$ of selected pairs and shortest distance in S.
$\varepsilon,\ L$	Relative tolerance ($0 < \varepsilon \leq 1$) and total binary input length, including rational accuracy.
Exchange and geometric rounding	
$C_j,\ r_j$	First j farthest-first centers and distance of center j to its nearest predecessor ($j \geq 2$). The proof uses $C = C_{2k}$ and $r = r_{2k}$.
$A,\ T_x,\ \Psi_A(z)$	Companions $S \setminus \{x\}$, centers at distance at least $r/2$ from all of A, and the interaction of z with A.
$H_m^{(p)},\ H_m$	Generalized harmonic sum $\sum_{j=1}^{m} j^{-p}$ and its value at $p = 1$. $\overline{H}$ denotes a computable upper bound.
$\delta,\ \sigma,\ \widehat{\sigma}$	Separation proved by exchange, any valid lower bound on optimal pair distances, and a rounding scale in $(\sigma/2, \sigma]$.
$q,\ h,\ o$	Fixed integer $q \geq \max\{1, s\}$, grid side $\varepsilon\widehat{\sigma}/(4qd)$, and common physical grid origin.
$Q,\ M,\ T$	Representatives, their number $M = \lvert Q\rvert$, and a budget of complete subset evaluations, to compare with $\binom{M}{k}$.
$\lambda,\ \eta$	Upper bound on doubling dimension and covering radius of the greedy representative net.
Bounds and experiments	
$U,\ \overline{U}$	In the certificate comparison: energy of the k-center incumbent and an upper bound on that energy.
$B_X,\ B_H$	Upper bounds on a single optimal pair contribution: exchange and hybrid, with $B_H = \min\{\overline{U}, B_X\}$ in the plane.
$V,\ U_t$	Exact reduced optimum and original energy after t within-cell refinement sweeps ($U_0 = V$). The certified interval is $[V/(1+\varepsilon), U_t]$.
$c,\ G$	Number of clusters and number of positive-probability cells in the fixed-grid occupancy model.
PC, PK, I, X, H	Pair-count, incumbent-packing, evaluated-incumbent, exchange and hybrid certificates. M_I, M_X, M_H are the corresponding grid occupancies for I, X and H; T is the enumeration budget.

B Proof details and implementation

We prove the planar sharpness result, then give the arithmetic and search details needed to implement the geometric bounds. The last subsection treats feasible domains that are not given as lists of sites.

B.1 Why the planar separation scale is sharp

The critical energy in a square is of order $m^2 \log m$ on the unit scale; compare the continuous asymptotic of Hardin and Saff [7, Theorem 2.1, (8)]. For the finite exchange construction we need just the following uniform lower bound, which has an elementary proof. All pair sums here are unordered.

Lemma B.1 (Energy forced inside a square)**.** *There is an absolute $c > 0$ such that every $m \geq 4$ distinct points in a square of side $a > 0$ have inverse-square energy at least $c\, m^2 \log(m)/a^2$.*

Proof. At level j, subdivide the square into 4^j congruent subsquares, assigning boundary points consistently so that the partitions are nested. If m_C points are in cell C, the number of same-cell pairs satisfies

$$N_j = \tfrac{1}{2}\big(\textstyle\sum_C m_C^2 - m\big) \;\geq\; \tfrac{1}{2}(m^2/4^j - m) \;\geq\; m^2/(4 \cdot 4^j), \qquad 0 \leq j \leq J := \lfloor \log_4(m/2) \rfloor.$$

For any pair p, q, let ℓ be its last same-cell level through J. Its squared distance is at most $2a^2/4^\ell$. Thus its total weight in the sum of the level counts is $\sum_{j=0}^{\ell} 4^j/a^2 \leq (4/3)4^\ell/a^2 \leq (8/3)\|p-q\|^{-2}$. Summing over pairs yields $E_2 \geq (3/8)\sum_{j=0}^{J} 4^j N_j/a^2 \geq 3(J+1)m^2/(32a^2)$. □

Proof of Theorem 3.3. Let $a = 2\lceil\sqrt{k}\rceil$, and take the lattice square $G = \{0, \ldots, a-1\}^2$. Adjoin $u = (k^2, 0)$ and $v = (k^2 + t, 0)$, where t is a dyadic rational in $[A/\sqrt{\log k}, 2A/\sqrt{\log k}]$. The absolute constant A is chosen below. For all sufficiently large k, $t < 1$ and the remote sites are at distance $\Omega(k^2)$ from G. There are at least $4k$ lattice sites. Until $2k$ centers have been chosen some untouched lattice site is at distance at least one from every center; hence $r_{2k} \geq 1$. At least $2k - 2$ centers are in G. Partition the lattice sites into $\lceil\sqrt{k}\rceil^2$ disjoint blocks of 2 by 2 sites. For all sufficiently large k there are fewer blocks than lattice centers. Two of these r_{2k}-separated centers lie in the same block, so $r_{2k} \leq \sqrt{2}$. Neither argument uses the initial site or tie-breaking.

Consider a k-set omitting at least one of u, v. It selects $m \geq k - 1$ lattice sites. Their mean incident energy, counting only their mutual interactions, is $2E_2/m$. Lemma B.1 and $a^2 = O(k)$ give a lattice site with incident energy at least $c_0 \log k$ for an absolute $c_0 > 0$. Replacing it with an omitted remote point costs at most $t^{-2} + O(k^{-3})$: at most one interaction is across the remote pair, and each of at most $k - 1$ lattice interactions costs $O(k^{-4})$. Choose A so that $A^{-2} < c_0/2$. The replacement is strictly improving for every sufficiently large k. Every one-swap local optimum must therefore contain both remote points. All its other distances are at least one, so its minimum distance is exactly t. Since $1 \leq r_{2k} \leq \sqrt{2}$, the claimed order follows. The remote-pair term alone also gives incident energy $\Omega(r_{2k}^{-2} \log k)$ at each remote site, matching the order of Lemma 3.2 there. □

B.2 A rational harmonic surrogate and dyadic scale

For fixed even integer $s > 0$ and fixed integer dimension d, the harmonic sum $H_{2k}^{(s/d)}$ need not be rational. Put $m = 2k$ and $b = \lceil \log_2 m \rceil$, computed from integer bit lengths. Define the rational

upper surrogate

$$\overline{H} = \begin{cases} \dfrac{d}{d-s}\, 2^{\lceil b(d-s)/d \rceil}, & s < d, \\ 1 + b, & s = d, \\ \dfrac{s}{s-d}, & s > d. \end{cases} \tag{B.1}$$

It satisfies $H_m^{(s/d)} \le \overline{H} = O_{d,s}(H_m^{(s/d)})$. For $p = s/d < 1$, integral comparison gives

$$H_m^{(p)} \le \frac{m^{1-p}}{1-p} \le \frac{2^{\lceil b(1-p) \rceil}}{1-p}.$$

Conversely $H_m^{(p)} \ge m^{1-p}$, and $2^b < 2m$ bounds the ratio by a constant depending only on p. For $p = 1$, $H_m \le 1 + \ln m \le 1 + b$, while $H_m \ge \ln(m+1)$ gives the constant-factor converse. For $p > 1$, $H_m^{(p)} \le 1 + \int_1^\infty x^{-p}\, dx = p/(p-1)$ and $H_m^{(p)} \ge 1$. This proves all three cases. In the planar critical case, the exact rational H_{2k} may be used instead.

Set

$$B = \frac{4^s \overline{H}}{(r^2)^{s/2}}, \qquad \delta_0 = B^{-1/s}.$$

Then δ_0 is a valid optimum-separation lower bound, within a fixed factor of (3.5). Since s is fixed and even, B is positive rational. Write $B = A/D$ with positive integers A, D, and let $\mathrm{bits}(z) = \lfloor \log_2 z \rfloor + 1$ for a positive integer. Compute

$$e = \left\lfloor \frac{\mathrm{bits}(D) - \mathrm{bits}(A)}{s} \right\rfloor.$$

If $B 2^{es} > 1$, decrement e once. The resulting $\widehat{\delta} = 2^e$ obeys

$$B\widehat{\delta}^s \le 1 < B(2\widehat{\delta})^s, \qquad \delta_0/2 < \widehat{\delta} \le \delta_0.$$

For completeness, put $\ell = \mathrm{bits}(D) - \mathrm{bits}(A)$ and write $\ell = es + t$, $0 \le t < s$, for the initial exponent. Bit lengths imply

$$2^{-\ell-1} < B < 2^{-\ell+1}, \qquad 2^{-t-1} < B 2^{es} < 2^{1-t}.$$

Thus the initial scale is either valid or at most one dyadic step too large; its next larger step is strictly invalid. After a decrement, the previous value $B 2^{es} > 1$ supplies the required strict upper comparison. This proves the claimed constant-number comparison procedure.

All integers in this construction, including the grid indices, have polynomial bit length in L. The magnitude of e is polynomially bounded in the encoding length of B; shifts implement powers of two exactly. The surrogate is computed using fixed rational constants, integer ceilings, and shifts. Consequently preprocessing and enumeration have polynomial bit cost per stated operation, and no algorithm step treats an irrational boundary or an arbitrary-real floor as a bit-model primitive.

B.3 Nested grids and incumbent bounds

For each positive certificate B, let $\widehat{\sigma}(B)$ be the largest power of two satisfying $B\widehat{\sigma}(B)^2 \le 1$. If $B_1 \le B_2$, then $\widehat{\sigma}(B_1)/\widehat{\sigma}(B_2)$ is an integral power of two. With common accuracy and physical origin o, the cell width ratio is therefore an integer and each coarse half-open cell is a union of fine cells. Every occupied coarse cell contains an occupied fine cell, proving $M(B_1) \le M(B_2)$. It follows that the hybrid has $M_H = \min(M_I, M_X)$. Representative identities can change, so this

assertion is not an ordering of the returned energies. Origins chosen as different fractions of each method's width need not be common physical origins and do not support this nesting proof.

For integer coordinate numerators with common denominator D_0, write $q_{ij} = \|X_i - X_j\|^2$ and set $A = 2^{48}$. The integer sum

$$\widetilde{U} = A^{-1} \sum_{\{i,j\} \subset C_k} \left\lceil A D_0^2 / q_{ij} \right\rceil$$

satisfies $U \le \widetilde{U} < U + \binom{k}{2}/A$. Taking $\overline{U} = \min\{\widetilde{U}, B_{\mathrm{PK}}\}$ remains valid because both quantities bound U. This avoids summing fractions with large unrelated denominators in the larger-k occupancy study. All ceiling operations use integers. The small-instance tests independently verify that this bound dominates exact incumbent energy.

There is a useful endpoint check when comparing these certificates under $T = 10^7$. For a planar incumbent C_k at mutual distances at least r_k, draw an edge for each distance strictly below $\sqrt{2}\, r_k$. The straight-line edges cannot cross: in a convex quadrilateral some angle is at least 90°, and the diagonal opposite that angle has length at least $\sqrt{2}\, r_k$ by the cosine rule. Two crossing edges would be its diagonals. A short edge also cannot pass through a third site, since its two subsegments would each have length at least r_k. Thus there are at most $3k - 6$ short pairs for $k \ge 3$. Each short pair contributes at most $1/r_k^2$, and each other pair at most $1/(2r_k^2)$. Consequently

$$U \le \frac{N_k + 3k - 6}{2 r_k^2}.$$

At $k = 12$ the numerator after division by two is $48 < 16 H_{24}$. Together with the $k \le 11$ comparison and $M_X \ge 2k$ for larger k, this shows that, with the exact incumbent, exchange alone cannot newly admit a raw enumeration at this budget for any k on the $n > 2k$ branch. One may also cap a rounded incumbent by this planar bound; the reported benchmarks use the PC/PK cap described above and do not rely on this extra cap.

B.4 Cluster size, grid placement and the search budget

If each cluster is contained in a ball of radius ρ, its coordinate range on each axis is at most 2ρ. The extreme floor indices differ by at most $\lceil 2\rho/h \rceil$, so $M \le c(1 + \lceil 2\rho/h \rceil)^d$. When there are $M = k + t$ representatives, the enumeration count is $\binom{k+t}{t}$. For $t = 0$ this is one and for $t = 1$ it is $k + 1$. Even $M = 2k$ leaves $\binom{2k}{k}$ subsets, which grows exponentially in k. The implementation evaluates `math.comb(M,k)` before allocating reduced pair weights.

For a fixed grid and n independent draws, let $p_j > 0$ be the probability of a site falling in cell j, for $1 \le j \le G$. Cell j is empty with probability $(1 - p_j)^n$. Summing the probabilities that the cells are occupied gives

$$\mathbb{E}[M] = \sum_{j=1}^{G} \left[1 - (1 - p_j)^n \right].$$

For equal-probability cells this becomes $G[1 - (1 - 1/G)^n]$. At low density most sites have their own cell. As n increases at fixed G the expected count approaches G. The experiments compute occupancy directly because the certified width depends on the input and the dyadic uniform generator samples without replacement.

B.5 Exact reduced search and numerical safeguards

The package contains `anc/benchmarks/code/planar_riesz_epas_v2.py`, the standard-library rational reference, and `anc/benchmarks/code/epas_refine_v2.py`, an accelerated planar implementation for the bounded dyadic data used here. The reference handles arbitrary rational

coordinates and small-cardinality branches. The vectorized path is intended for validated planar arrays and the stated $2k$-cover regime; the small walkthrough additionally uses the proved $2k-2$ variant. Both use the last insertion distance to set the grid scale. Original points are grouped by cell and remain available after reduction.

Exact integer geometry. Write a candidate as X_i/D, where $X_i \in \mathbb{Z}^2$ and $D = 2^{20}$. The code first checks distinctness and that the sum of squared coordinate spans is at most 2^{53}. Squared pair distances in integer units can then be evaluated exactly in signed 64-bit arithmetic and converted exactly to binary64 when needed. They are positive for distinct candidates. If $hD = a/b$ in lowest terms and the grid origin is $(-h/2, -h/2)$, the cell key is

$$\left(\left\lfloor \frac{2bX_{i,1} + a}{2a} \right\rfloor, \left\lfloor \frac{2bX_{i,2} + a}{2a} \right\rfloor \right).$$

The code verifies the integer expression does not overflow; floor division handles negative coordinates. Sorting by the two keys and then input index retains exactly one original candidate per occupied cell. No rounded distance determines a center, cell, or certificate.

Exact reduced search. Let $w_{ij} \in \mathbb{Q}_{>0}$ be a representative pair weight and $A = 2^{48}$. The search uses integer lower weights $\ell_{ij} = \lfloor A w_{ij} \rfloor$. If S_{inc} is the best feasible set found so far, the pruning threshold is $U = \lceil A E_2(S_{\mathrm{inc}}) \rceil$. A deterministic greedy completion gives an initial feasible set. For a partial selection with lower cost c, let b_i be the lower attachment cost of available candidate i, and let q positions remain. If $q \geq 2$, assign

$$a_i = 2b_i + \text{the sum of the } q-1 \text{ smallest } \ell_{ij} \text{ over other available } j.$$

For any completion T, $|T| = q$, the sum of its charges is at most twice its remaining lower cost: each internal edge is counted twice and the attachment costs are doubled. Hence the sum of the q smallest charges over all candidates is also a lower bound on twice the remaining cost. Prune only when this bound plus $2c$ exceeds $2U$. For $q = 1$, use $c + \min_i b_i > U$. Positivity also permits discarding a candidate whose already forced lower cost exceeds U. Every surviving leaf is compared using exact rational weights; equality is resolved by the lexicographic input-index tuple. Conservative floors can weaken pruning but cannot remove a better or tied optimum. A timed-out traversal is never reported as a completed EPAS.

Certified potential filtering in refinement. One candidate potential is $D^2 \sum_j 1/q_j$, with exact positive integers $q_j \leq 2^{53}$. The common factor D^2 does not affect minimization. Under IEEE-754 binary64 round-to-nearest arithmetic, each division $1/q_j$ is enclosed by the adjacent representable numbers obtained with `nextafter`. After every positive addition, outward expansion encloses the accumulated exact sum. Values stay far from underflow. Let $[\ell_p, u_p]$ be these enclosures and $u_{\min} = \min_p u_p$. A true minimizer $p^\star$ satisfies $\ell_{p^\star} \leq u_{\min}$: its exact potential is no larger than that of a candidate attaining the smallest upper endpoint. Thus candidates with $\ell_p > u_{\min}$ can be discarded safely. All others are evaluated as exact fractions, together with the incumbent. Ties retain the incumbent; otherwise the smallest input index among strict minimizers is used. This is a filtering implementation of exact minimization, not floating-point acceptance with a tolerance.

Refinement cost and certificates. The selected cells and their visit order remain fixed. An update changes only the incident potential and updates the exact total by that difference. The code checks the exact total again after each sweep, asserts nonincrease after every update, and verifies the selected cell identifiers and distinctness. A no-improvement sweep is a fixed point for this deterministic within-cell rule. Requested later prefixes reuse the result without additional

work. Such early stopping is valid for fixed-t output and does not establish a worst-case bound for unrestricted convergence. The posterior interval retains the base lower endpoint $V/(1+\varepsilon)$ while replacing the upper endpoint by the refined energy.

B.6 Polygon witnesses and their coordinates

Corollary 4.1 assumes that both geometric primitives can be computed. On any compact feasible set with at least $2k$ distinct points, continuous farthest-first traversal exists: the distance to a nonempty finite set is continuous and attains its maximum. The maximum stays positive through the $2k$th insertion, and the maxima are nonincreasing. Hence the last insertion distance gives separation and coverage. An existence argument alone gives no procedure for locating these maximizers on an arbitrarily represented set.

For a filled simple polygon with m vertices, start at a polygon vertex. With j sites Chew and Drysdale give $O(mj + j\log(2j))$ geometric operations, or $O(mj)$ with the nearest-site Voronoi diagram supplied [16]. Summing gives $O(mk^2 + k^2\log(2k))$ operations for the certificate. The feasible location is the center of the empty circle. A boundary site is eligible even when its circle extends outside the polygon. These bounds are for a simple polygon with its boundary included.

Write G for the number of cell keys visited in the cover boxes, counting duplicates. For a box $[l_i, u_i]$ on axis i and grid origin o, enumerate the integers from $\lfloor (l_i - o_i)/h \rfloor$ through $\lfloor (u_i - o_i)/h \rfloor$, inclusively. Take their Cartesian product, sort and deduplicate, then test each key. With exact-real box arithmetic, $G \le 2k(2 + 2r/h)^d$. In a rational implementation one may enclose r by a dyadic $r \le r' < 2r$ and use r' just for the boxes. This changes the count by at most a constant factor at fixed dimension.

Triangulate the polygon once, for example by ear clipping in $O(m^3)$ elementary operations after removing redundant collinear boundary vertices. Clip each closed triangle against the closure of a queried square. If the result is nonempty, average its distinct vertices, including the cases of a single point or segment. This point lies strictly below each upper coordinate bound unless the whole clipped set lies in the corresponding upper face. Thus it belongs to the half-open square exactly when a witness exists in that triangle. The lower faces are included. Stop at the first witness across all triangles. For example the triangle with vertices $(0,0), (1,0), (0,1)$ intersects $[1,2) \times [0,1)$ at $(1,0)$ and has empty intersection with $[-1,0) \times [0,1)$. This direct procedure uses $O(m^3 + G\log(2G) + mG)$ operations including triangulation.

For rational vertices and planar $s = 2$, use the dyadic scale of Theorem 3.1. A farthest point can be chosen among polygon vertices, bisector–boundary intersections and Voronoi vertices: on a clipped nearest-site cell the squared distance to that site is convex and attains a maximum at a vertex after triangulating the clipped region. Degenerate ties admit the same choice of a vertex, including endpoints of overlapping edges. Every selected coordinate therefore solves a rational linear system of order at most two using original vertices and at most three existing sites. With B bits per current coordinate, a constant number of rational arithmetic operations gives $O(B + L)$ bits for the new coordinate, where L is polygon and accuracy encoding length. Iterating for $2k$ steps permits $C^{O(k)}(L+1)$ bits, sufficient for polynomial bit cost in L at fixed parameters. For an elementary implementation of the certificate with this bit accounting, enumerate the $O(j^3 + mj^2 + m)$ candidate intersections from the original features of the current diagram and boundary; test polygon membership and score nearest-site squared distance in $O(m + j)$ operations per candidate. Extraneous feasible intersections cannot exceed the true maximum. This slower construction avoids assumptions about bit growth inside a particular Voronoi implementation. Dyadic cell keys, triangle clipping and the rational energy evaluations obey the same parameter-dependent bit bound. No polygon complexity enters the exponential subset-enumeration count.

Metric arithmetic. In Theorem 4.2 an arbitrary fixed real λ is an upper bound and the formulas use exact powers. For an integer dimension bound and rational distance-oracle outputs, a fixed positive integer s permits the rational harmonic surrogate in (3.6) with d replaced by that bound. Rounding the resulting separation scale down to a dyadic number loses at most another factor of two in the net radius. Bit costs then depend on the oracle output lengths and the cost of answering a query.

C Worked examples: seeing the grid at three densities

The ten-site instance lets us identify every feasible replacement. The two instances with $n = 100$ use the same initial sites and illustrate how the placement of the other candidates changes grid occupancy. All three use $k = 5$.

C.1 Five of ten: from cover to selection

Figures 3–4 illustrate $d = s = 2$, $n = 10$, $k = 5$, and $\varepsilon = 1/2$. The $2k$ implementation enumerates directly when $n = 2k$; the $2k - 2$ variant below makes both covering and rounding visible even on this small input.

Proposition C.1 (Two fewer covering centers)**.** *For $k \geq 2$, the planar construction remains valid with $m = 2k - 2$ farthest-first centers, $H = H_m$, and $B = 16H/r^2$, using direct enumeration when $n \leq m$. It has the same approximation guarantee and asymptotic bounds as Corollary 3.5.*

Proof. For any selected x, its $k - 1$ companions exclude at most $k - 1$ centers, so at least $m - (k - 1) = k - 1$ free centers remain. In Lemma 3.2, replace $2k$ by m: the averaging factor $(k - 1)/|T_x|$ is still at most one. The packing sum is bounded by $16r^{-2}H_m$. Farthest-first still gives an r-separated cover, and the grid, injectivity, and bit arguments are unchanged. Also $m = O(k)$, $H_m = O(\log(2k))$, and $n \leq m$ implies $\binom{n}{k} \leq 4^k$. This changes the number of covering centers by a constant and preserves the approximation principle. □

Implementation. The program `planar_riesz_epas.py` uses Python's standard-library `Fraction` throughout. Default `paper` mode follows the $2k$ construction; `compact` uses Proposition C.1. It maintains nearest squared distances, constructs dyadic scales by binary lengths, sorts integer cell keys, and tabulates only representative weights. Input order breaks ties and chooses cell representatives. A translated global grid has the same proof, with keys $\lfloor (p - o)/h \rfloor$. The solver optimizes over the representatives. A separate exhaustive search provides the full-input optimum for comparison.

The ten input points, in order, are as follows, with $t = 1/512$:

$$\begin{aligned} &p_1 = (0, 0), \quad p_2 = (4, 0), \quad p_3 = (7, 2), \quad p_4 = (7, 6), \quad p_5 = (3, 8), \\ &p_6 = (-1, 6), \quad p_7 = (-3, 3), \quad p_8 = (2, 3), \quad p_9 = (-t, -t), \quad p_{10} = (3, 8 + t), \end{aligned}$$

The two deliberate near pairs require magnification. Starting at p_1, the trace gives

$$\mathcal{C} = (p_1, p_4, p_6, p_{10}, p_2, p_3, p_7, p_8), \qquad r^2 = 13, \quad H_8 = 761/280, \quad B = 1522/455.$$

The insertion squared distances are $85, 37, 5244929/262144, 16, 13, 13, 13$. The final covering radius is actually $\sqrt{2}/512$, but the proof uses the *last insertion radius* $r = \sqrt{13}$. The dyadic choice $\widehat{\delta} = 1/2$ satisfies $B\widehat{\delta}^2 \leq 1 < B(2\widehat{\delta})^2$, giving $h = 1/64$. With origin $o = (-h/2, -h/2)$, exactly two cells contain pairs; the first-point rule leaves $Q = \{p_1, \ldots, p_8\}$.

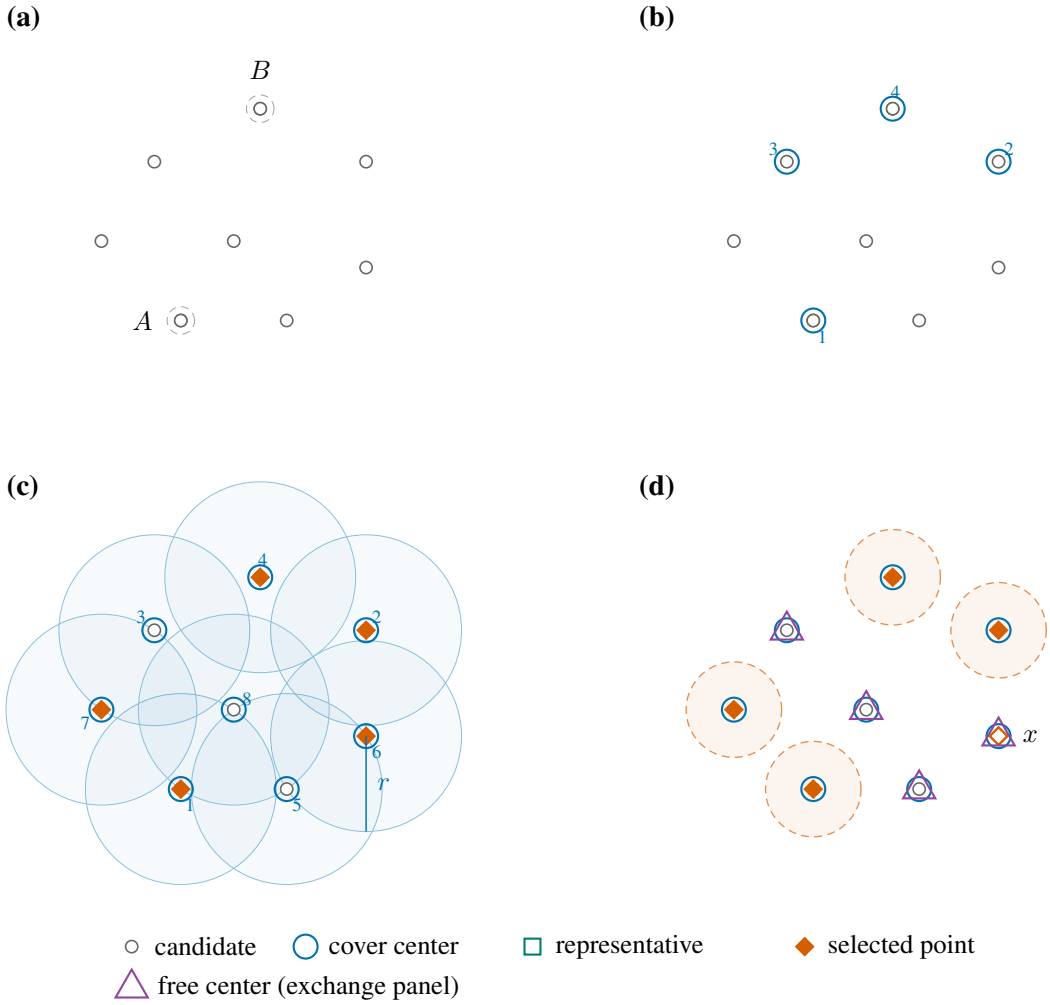


Figure 3: Cover and exchange for ten sites, $k = 5$ and $\varepsilon = 1/2$. (a) The pairs at A and B are magnified in Figure 4. (b) Numerals give the order of the first four farthest-first centers. (c) Blue rings mark the eight centers of the $2k - 2$ cover, with disks of radius $r = \sqrt{13}$. Orange diamonds mark the optimum $\{p_3, p_4, p_7, p_9, p_{10}\}$. (d) To replace $x = p_3$ (outlined diamond), exclude the open radius-$r/2$ disks of its four companions. The four free centers p_2, p_3, p_6, p_8 are marked by purple triangles. Their average replacement cost bounds the energy incident to x.

Stage	Exact result	Role
Cover	8 centers, $r^2 = 13$	Certifies the rounding scale
Grid	$10 \to 8$ candidates	Merges only the two near pairs
Representative search	$\binom{8}{5} = 56$ subsets	Produces the returned solution
Independent audit	$\binom{10}{5} = 252$ subsets	Checks the unknown original optimum
Certificate	$[2V/3, V]$	Valid without the exhaustive audit

The returned set is $S_Q = \{p_1, p_3, p_4, p_5, p_7\}$, with

$$V = \frac{110421302865713}{413429256648720} \approx 0.267086330.$$

The independent verifier finds the unique original optimum $S^\star = \{p_3, p_4, p_7, p_9, p_{10}\}$, of energy 0.267019408 approximately. Its cellwise replacement equals S_Q here, giving the nonzero-error ratio $1.000250627 < 1.5$. This equality of the replacement and returned set is not needed in general. The $2k$ mode instead takes the exact 252-subset shortcut.

From `anc/walkthroughs/`, run the independent audit with

```
python epas_gallery_run.py
```

This runs both the compact and standard modes, then enumerates the original ten-site problem independently. It writes `epas_gallery_results.json` and checks 78 additional cases covering cardinalities, coordinate signs, scales, translations and accuracy endpoints, as well as four invalid inputs.

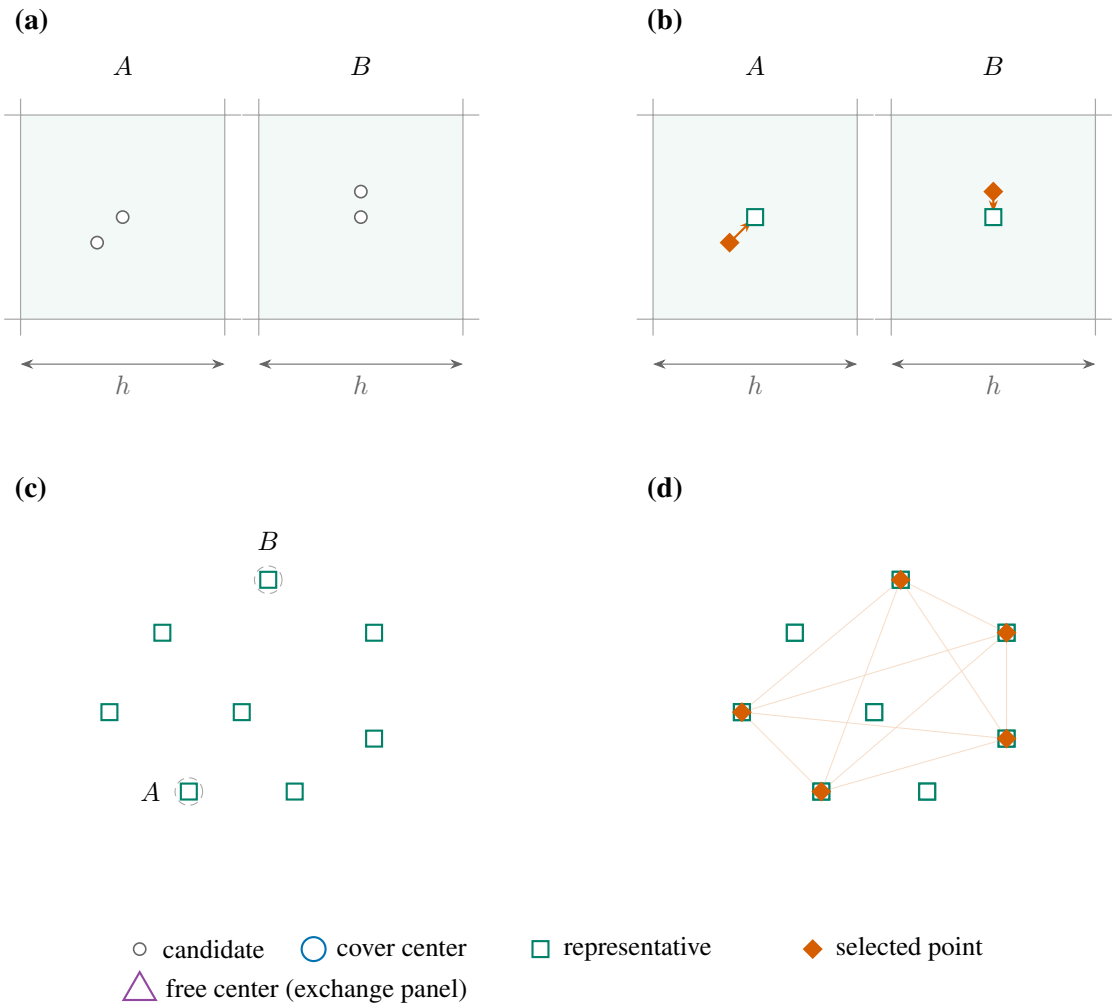


Figure 4: Rounding and search on the same ten sites. (a) The two magnified cells have side $h = 1/64$ and belong to the grid with origin $(-1/128, -1/128)$. Cell A contains p_1, p_9 and cell B contains p_5, p_{10}. (b) The first site in each cell is retained (green square). Two sites of the optimum (orange diamonds) move under rounding, $p_9 \mapsto p_1$ and $p_{10} \mapsto p_5$. The other three stay fixed. (c) The grid retains eight representatives. (d) Searching their $\binom{8}{5} = 56$ five-site subsets returns $\{p_1, p_3, p_4, p_5, p_7\}$. Segments show its ten interacting pairs. The certified interval is $[2V/3, V]$. An independent exhaustive search gives $V/\operatorname{OPT} \approx 1.000250627$.

C.2 A hundred sites spread across the same cover

Keep the first ten candidates and append 90 points spread across the covered region, preserving the eight-center traversal, $k = 5$, $d = s = 2$, and $\varepsilon = 1/2$. Figure 5 plots each candidate at its generated coordinates.

Let $c_1, \dots, c_8$ be the ordered centers above, and r_j^2 their squared insertion distances. Form the finite pool of points

$$q = (a/4, b/4), \qquad -28 \le a \le 44, \quad -16 \le b \le 48, \qquad a, b \in \mathbb{Z},$$

excluding existing candidates, and retain only those satisfying

$$\min_{1 \le \ell < j} \|q - c_\ell\|_2^2 \le r_j^2 \quad \text{for every } j = 2, \dots, 8.$$

Exactly 1402 pool points survive. Append 90 points, each maximizing its minimum squared distance to all candidates already present; lexicographic coordinates break ties. This defines the input. The prefix inequalities prevent added points from outranking prescribed centers; equality favors the smaller original index. Induction preserves the traversal. The admissible region is contained in the covering-disk union; both the last insertion radius and final cover radius are $\sqrt{13}$.

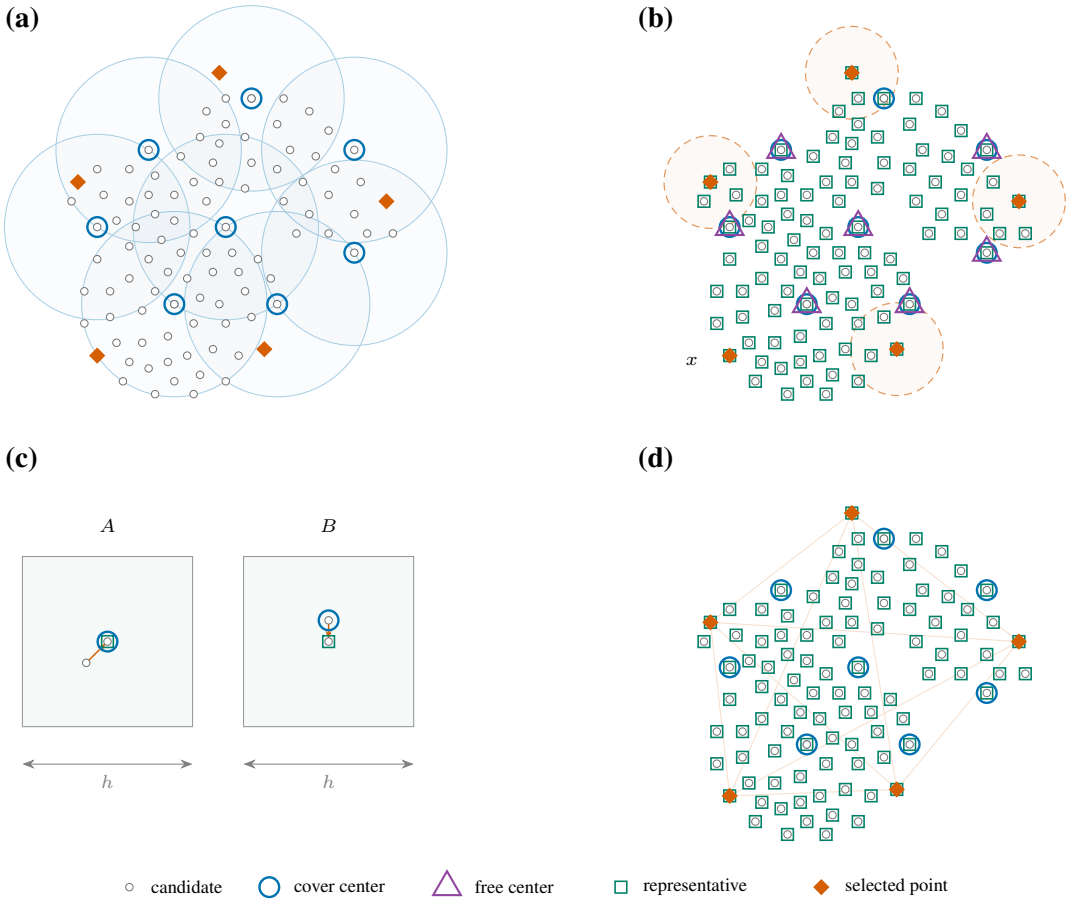


Figure 5: One hundred sites with $k = 5$, eight cover centers, $r = \sqrt{13}$ and $h = 1/64$. (a) The disks cover all candidates. Orange diamonds mark the five sites of the independently verified optimum. (b) Green squares mark the 98 representatives and blue rings the eight centers. For $x = p_{12}$, the four dashed disks about its companions exclude only p_{10}, leaving seven free centers (purple triangles). (c) The only cells containing two sites are $\{p_1, p_9\}$ and $\{p_5, p_{10}\}$, shown at a common magnification. All five optimal sites are retained. (d) The selected sites and their ten pair interactions.

The potential bound $B = 1522/455$, dyadic scale $1/2$, grid side $h = 1/64$, and origin $(-1/128, -1/128)$ are unchanged. Recomputing all cell keys yields 96 singleton cells and the two cells $\{p_1, p_9\}$ and $\{p_5, p_{10}\}$. Thus

$$Q = \{p_1, \dots, p_8, p_{11}, \dots, p_{100}\}, \qquad |Q| = 98.$$

The standard uniform grid therefore gives little compression on this spread-out input. A fixed cover does not imply a fixed occupied-cell count. The comparison is

n	k	Original k-subsets	$\lvert Q\rvert$	Representative k-subsets
10	5	252	8	56
100	5	75 287 520	98	67 910 864

These are compact-EPAS search-space sizes. The larger run tabulates 4753 representative weights and uses exact pruning instead of visiting every subset; no measured speedup is asserted.

Selection and certificate. Exact optimization on Q returns

$$S_Q = \{p_{12}, p_{19}, p_{30}, p_{33}, p_{87}\}, \qquad V = \frac{190915454332445247907150 84}{138473515345884275211104385} \approx 0.137871458.$$

The EPAS certificate is $[2V/3, V]$. A separate exact search on *all 100 original candidates* returns the same optimum, certifying $[V, V]$ and error $V/\operatorname{OPT}_{100} - 1 = 0$ for this instance. This audit does not construct the grid or improve the scheme's worst-case 1.5 guarantee or asymptotic search bound.

For the exchange illustration choose $x = p_{12}$. The seven free centers are

$$T_x = \{p_1, p_2, p_3, p_4, p_6, p_7, p_8\};$$

only p_{10} lies within distance strictly less than $r/2$ of one of $p_{19}, p_{30}, p_{33}, p_{87}$. In contrast to the ten-point example, no selected optimum point is a cover center. Free centers are determined by these exclusion disks, not by being outside S_Q. The figure plots all centers even when excluded and marks free ones by a second, distinct symbol.

Certified exact pruning. The optional `-search branch_bound` routine changes only the exact search on the retained set, leaving the EPAS geometry and default enumerator intact. For rational weights $w_{ij} > 0$, set $K = 2^{48}$ and $a_{ij} = \lfloor K w_{ij} \rfloor$. Maintain an exactly evaluated feasible incumbent of energy U. In a node with partial selection A, admissible remaining indices R, and $q = k - |A|$ still to select, put

$$c = \sum_{\{i,j\} \subseteq A} a_{ij}, \qquad b_i = \sum_{j \in A} a_{ij} \quad (i \in R).$$

For $q \geq 2$, let t_i sum the $q-1$ smallest values a_{ij} over $j \in R \setminus \{i\}$ and let D sum the q smallest values $2b_i + t_i$ over $i \in R$. For any completion $T \subseteq R$, $|T| = q$,

$$2KE_2(A \cup T) \;\geq\; 2c + \sum_{i \in T} (2b_i + t_i) \;\geq\; 2c + D.$$

Indeed the attachment terms are charged twice, while each internal completion edge is charged once at each endpoint; choosing the cheapest available incident edges only lowers these charges. Consequently $2c + D > 2\lceil KU \rceil$ safely discards the node. For $q = 1$, use $c + \min_{i \in R} b_i > \lceil KU \rceil$. Positivity also allows a candidate to be discarded if its partial lower cost already exceeds $\lceil KU \rceil$. The recursion follows increasing indices, so every remaining subset is either visited or safely pruned. Complete surviving selections and incumbent updates use exact rational energy, not rounded weights. Thus termination certifies the exact optimum; the integer scaling affects pruning strength, not correctness. Greedy completions supply feasible initial incumbents only.

The representative search visits $29\,539$ partial-tree nodes, and the separate all-candidate audit visits $32\,039$; these include partial selections and are not full-subset enumeration counts. Both

searches use the same exact routine on different domains. Independent exhaustive rational minimization validates the pruning implementation on 111 weighted test cases totaling $12\,009$ subsets, including tiny weights below $1/K$ and tied optima. The ten-point audit and its 78 additional geometric cases also pass.

From `anc/walkthroughs/`, run the following three commands. The first constructs the spread instance, solves it, and audits the full-input optimum; the second checks the pruning rule against exhaustive search; the third draws the figure.

```
python epas_100_run.py
python check_exact_subset_search.py
python render_epas_spread.py
```

JSON records the full rational input and trace; assertions verify the traversal, geometry, 98 representatives, and equality of both optima. Maximum replacement squared distance is $1/131072$; all added points represent themselves.

C.3 A hundred sites in tight groups

The five-of-ten and spread-out hundred-point constructions above determine the coordinates and their original ordering. Keep the original ten points. From the 90 added spread-out candidates, admit an anchor p only if all five offsets p, $p \pm (1/512, 0)$, $p \pm (0, 1/512)$ satisfy every original eight-center traversal inequality strictly. Starting with the original ten points, choose 18 such anchors successively by maximum distance to the already chosen points, breaking ties lexicographically. Append the five offsets in that fixed order for each anchor. This yields exactly 100 distinct dyadic candidates; geometrically they represent 26 local groups, with 18 five-site cities, two two-site groups, and six singleton sites. No geographic dataset or demand model is asserted.

The strict inequalities preserve the first eight farthest-first choices. The full ten-center traversal next selects p_{15} and p_{18}. The rational implementation gives $h = 1/64$ and $M = 26$. The representative optimum is $\{p_5, p_{16}, p_{41}, p_{51}, p_{61}\}$ with energy $V \approx 0.159096904279$. An exact search over all candidates gives $\{p_{10}, p_{18}, p_{42}, p_{55}, p_{63}\}$, and $V/\,\mathrm{OPT} \approx 1.000596624154$. Fixed-cell refinement returns this original optimum. The code records exact fractions for all energies and checks the free-center set and preservation of the first eight centers. The initial certificate is $[V/1.5, V]$; after refinement its upper endpoint can be reduced to the refined energy. Independent exact optimization is an audit, not a step of the EPAS.

From `anc/walkthroughs/`, run `python compressed_hundred_v2.py`. The inputs, exact energies and audit are recorded in `compressed_hundred_v2.json`. Compare the 26 occupied cells here with 98 in the spread-out construction of Appendix C.2.

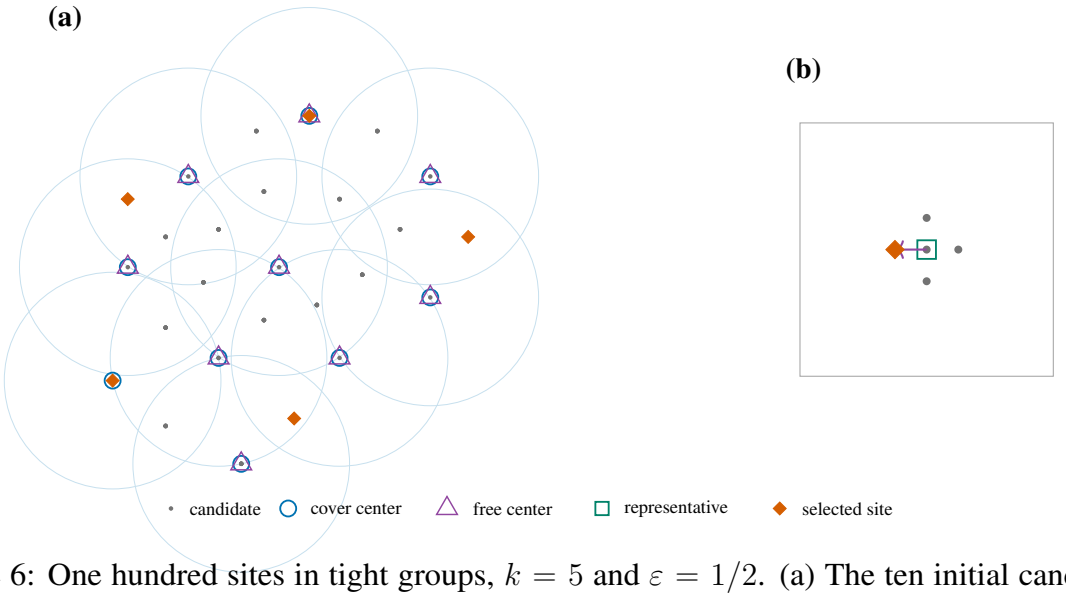


Figure 6: One hundred sites in tight groups, $k = 5$ and $\varepsilon = 1/2$. (a) The ten initial candidates and 18 groups of five share a cover with ten farthest-first centers (blue). Purple triangles mark the nine centers free to replace p_{10} in the optimum (orange). (b) A magnified cell of side $h = 1/64$ contains five candidates. Refinement moves its representative p_{16} to p_{18}. Across the input $M = 26$. The reduced solution is 0.05966% above the full optimum, and the refined solution attains it. The approximation guarantee is a factor of 1.5.

D Reproducing the Swiss settlement experiment

Choose ten of 5,403 prescribed sites in the Swiss LV95 plane (EPSG:2056). For integer metre coordinates we sum the 45 unordered inverse squared distances in m^{-2}. This is a facility *dispersion* example: there are no demand weights, travel times, or service-area constraints. Figure 1 shows the hybrid solution. The source package includes the displaced sites, projected outline, solver and exact reference results in `anc/swiss/`. The optimization can be repeated offline without downloading any original address data.

Where the candidates come from. The clouds follow the built-up areas of the 26 cantonal seats and ten rural settlements. We used swisstopo's 5 November 2026 building-address directory and the detailed January 2026 national polygon [20, 21]. Within each source locality we combined occupied 200 m squares by eight-neighbor adjacency and kept the component with the most distinct current address positions. Giving each occupied hectare equal weight, we stratified its centers into 50 groups by farthest-first initialization and Lloyd iterations, then used the address nearest each fitted center. The rural pool excluded municipalities already represented and required 100–2,000 residents, at most 50 residents per square kilometre, and at least 50 occupied hectares in the component of the municipality's most represented postal locality. We included the eligible settlement nearest the national area centroid, then chose nine further settlements successively by their distance from existing sites and settlement anchors. One site in Müstair represents the eastern border region. Thus a central settlement was deliberately available to the optimizer; selecting it was optional.

All 1,801 seed positions were displaced by 35–90 m, rounded to whole metres, and checked against the national boundary and the directory. The released, fixed seed cloud contains no address identifiers. To increase local density, we generated two neighbors of each seed in the 36 settlement clouds and a third for two seeds drawn without replacement, leaving Müstair isolated. Proposals were uniform by area in a 75–250 m annulus around their seed before rounding. We rejected

proposals outside the annulus or country, within 15 m of another accepted site, or within 15 m of any directory position. All 5,403 resulting coordinates are distinct, inside Switzerland and at least 15 m from the 3,307,712 directory positions checked. These are synthetic candidates, not actual building addresses. The generator uses neither the EPAS grid nor the computed facility locations.

From cover to representatives. Set $d = s = 2$, $k = 10$ and $\varepsilon = 1$. Start farthest-first traversal at the first CSV row, breaking ties by input index. The twentieth insertion has radius $r = 39.558$ km; after that insertion the covering radius is 37.954 km. Use the exact values of $B_X = 16H_{20}/r^2$ and of the energy U of the first ten centers, with r expressed in metres. Numerically $B_X = 3.67857 \times 10^{-8}$ and $U = 3.42232 \times 10^{-9}$. The hybrid uses $B_H = \min(B_X, U)$. For each certificate take the largest dyadic power z with $Bz^2 \le 1$ and set $h = z/16$. The grids have the common physical origin $(-128, -128)$ m. For the half-open cells, the key of (x, y) is $(\lfloor (x + 128)/h \rfloor, \lfloor (y + 128)/h \rfloor)$; the first CSV row in each occupied cell is its representative.

Table 5: Two certificates on the same 5,403 sites. Gaps are $100(E/\,\mathrm{OPT} - 1)$ relative to the exact full-input optimum. Moves count accepted replacements within the ten selected cells.

Certificate	h (m)	Retained	Reduced gap (%)	Moves	Refined gap (%)
Exchange	256	3,365	0.10125	3	0.09449
Hybrid	1,024	706	0.32553	8	0

Reading the Hasliberg panel. Each of its eight rings is the first site in one occupied hybrid cell; the 150 candidates occupy only these eight cells. The highlighted cell has lower left corner $(2\,657\,152, 1\,178\,496)$ m and contains nine candidates. The reduced solution uses P1320 at $(2\,657\,976, 1\,178\,623)$; refinement moves to P4949 at $(2\,657\,180, 1\,178\,604)$, holding the other nine facilities fixed for that move. Each sweep visits the reduced solution's cells in index order, minimizes the exact sum of the nine incident pair costs, and accepts only strict decreases; the runner stops at a sweep without a change or after ten sweeps. Eight accepted moves give exactly the ten sites found by the full audit: P0083 (Appenzell), P1048 (Bellinzona), P1301 (Müstair), P2602 (Schaffhausen), P3352 (Brusio), P3867 (Basel), P3980 (Genève), P4799 (Sion), P4949 (Hasliberg Wasserwendi), and P5312 (La Côte-aux-Fées). Hasliberg is 5.87 km from Switzerland's area centroid. Excluding every candidate within 30 km of that centroid raises the exact optimum by 1.57265%, so every optimal ten-set contains a central site under this definition.

What is certified. Let V be the exact representative optimum and U_{ref} the refined feasible energy. The hybrid gives $V = 2.8215031282 \times 10^{-9}$ and $U_{\mathrm{ref}} = 2.8123482117 \times 10^{-9}$ to the displayed precision. Before any full search the EPAS certifies $V/2 \le \mathrm{OPT} \le U_{\mathrm{ref}}$; the factor two concerns total energy, not individual distances. A separate exact full search proves $U_{\mathrm{ref}} = \mathrm{OPT}$ here. The hybrid has $\binom{706}{10} \approx 7.95 \times 10^{21}$ reduced subsets. On the representatives, the initial locality bound leaves 69 sets of ten localities for further exact search. The finer exchange grid gives a smaller initial gap, but its allowed replacements differ and its refinement does not reach the full optimum (Table 5). Exact recovery is not a refinement guarantee.

How the independent full-input optimum was computed. For each search domain, use a feasible ten-set as an upper bound. Check that the reciprocal squared diameter of every nonsingleton locality exceeds this bound. Even the smallest such pair bound on the full input is $1/135429557\ \mathrm{m}^{-2}$. Consequently an optimum uses at most one site per locality. During enumeration of the $\binom{37}{10} = 348{,}330{,}136$ possible locality products, bound each edge from below by the reciprocal of its maximum squared distance, rounded down at scale 2^{80}. For a prefix still needing t localities, each

possible new locality must pay its edges to the prefix and at least half the sum of its $t-1$ cheapest remaining edges when these charges are summed over new localities. The t smallest such charges bound the completion. Only 91 full-input products survive this stage.

Within a surviving product, round pair weights down at scale 2^{60}. For an integer pair matrix $W_{ij}(a,b)$, take each row minimum $u_i(a)$, then the column minimum of the residual $u_j(b) = \min_a(W_{ij}(a,b) - u_i(a))$. Since $u_i(a) + u_j(b) \le W_{ij}(a,b)$, summing the unary terms over pairs and minimizing separately over each domain gives a lower bound. A candidate's excess above its domain minimum can also eliminate that candidate. Bounds prune only above the scaled incumbent's ceiling. Domains that remain are split along their widest coordinate; surviving singleton products are compared as exact fractions. Floating-point initial solutions supply feasible bounds only. The code separately checks a 14-site restriction against exhaustive rational enumeration.

Reproduction. From `anc/swiss/`, run `python run.py` with NumPy, then `python make_figure.py`; no network access or source geodata are needed. The first command verifies the frozen CSV's SHA-256 and computes the cover, both grids, exact representative optima, accepted moves, full optimum, and central-exclusion audit. The second generates the inline TikZ from those results and the projected outline. The result JSON records site indices (zero-based data rows), exact fractions and search counts. The CSV uses IDs P0001 through P5403. To regenerate the densification itself, run `python make_dense.py` with the supplied displaced seed cloud and the two official source files specified in the README; the PCG64 seed is 20261107. The original displacement is frozen to avoid releasing an address correspondence. Source hashes and validation distances are in `generation.json`.

E Experimental protocols and results

The Swiss instance permits an independent full-input optimum. The synthetic studies answer a different question: as size, geometry and accuracy change, when is the reduced problem small enough to solve? We retain all rejected and timed-out cases in the occupancy statistics; only completed searches can support claims about refinement gains.

E.1 Fixed-grid refinement ablation

The exchange-grid study varies input size, group count and width, selected cardinality, accuracy, and grid translation. It comprises 156 runs: 153 at $\varepsilon = 1/2$ and 3 at $\varepsilon = 1$. We attempt exact reduced search only for $M \le 300$ and $\binom{M}{k} \le 10^{10}$, with a 30-second cap. Of these runs, 84 completed, 3 timed out and 69 exceeded a size limit. Only completed runs enter the refinement table.

For each sweep cap we take the corresponding prefix of the same trace, including the time to confirm a fixed point when the cap permits. Total time is preprocessing plus reduced search plus refinement; it excludes input generation and separate optimum or heuristic audits. Medians are taken after forming each run's total. Median preprocessing and reduced-search times were 2.99 and 3.78 ms, respectively. The gain is $100(V - U_t)/V$ for that run, with V its exact reduced optimum.

Sweep cap	Median E_s	Median gain (%)	Extra (ms)	Total (ms)
0	2.913211	0.00000	0.00	31.95
1	2.908612	0.05231	2.94	40.51
2	2.908612	0.05231	5.48	50.49
3	2.908612	0.05231	5.48	50.49
5	2.908612	0.05231	5.48	50.49
10	2.908612	0.05231	5.48	50.49

Of the 84 completed runs, 78 first reached their final recorded energy in sweep one and 6 in sweep two. Nine additional checks use the same generator and seeds $17, 29, 43$ at $\varepsilon = 1/2$. For $n = 20$, $k = 4$, uniform inputs kept all 20 sites; eight tight groups kept eight sites for every seed. Exhaustive search on the six original inputs confirmed that 5 refined solutions attained the optimum. For $n = 1000$, $k = 5$, ten tight groups kept ten representatives, with refinement gains of $0.0238 - -0.0587\%$. The exact energies and traces are in `anc/benchmarks/`; `anc/geometry/` contains `check_geometry_v3.py` for the polygon traversal, cell witnesses and twelve metric instances.

Accuracy and cluster geometry. The lattice spacing is one; ρ is the cluster half-width in these units. At $n = 10^4$, $k = 5$, $c = 10$ and $\rho = 1/1024$, both $\varepsilon = 1/2$ and 1 retained ten exchange representatives on every seed: doubling the permissible cell width did not merge any more groups. At $n = 10^5$ and $\varepsilon = 1/2$, increasing ρ from $1/4096$ to $1/1024$ and $1/256$ gave 10, 10, and 90 representatives, respectively. At $\rho = 1/4096$ and $c = 5$, the $2k$-center traversal must enter groups more than once; the small final insertion radius left 89,215–89,315 representatives. At $c = 10, 25, 50$ the same rule retained one site per group on all three seeds. These exchange counts use origin $(-h/2, -h/2)$ at the exchange width.

On the same five-group inputs, we can see exactly when M reaches k. The five unit-lattice centers each carry a square cloud of half-width $1/4096$. The hybrid gives $h_H = 1/128$ on all three seeds. At origin $(0, 0)$ every group straddles two grid lines and occupies four cells. Translating the origin to $(-1/256, -1/256)$ places every group inside one cell. The $2k$-center exchange width is $h_X = 1/524288$. The translation is an integer multiple of h_X along each axis, so its occupied-cell count is identical at both origins. Table 6 gives the resulting search sizes. Run `python code/occupancy.py` from `anc/benchmarks/` to reproduce the counts and the unique five-site reduced solutions.

Table 6: Search size known at the end of preprocessing. The synthetic cases use $k = 5$, $\varepsilon = 1/2$ and three seeds. The Swiss case uses $k = 10$ and $\varepsilon = 1$. On the uniform square the half-cell offset and $h = 1/256$ give $1{,}025^2$ cells meeting $[0, 4)^2$. The last column counts all reduced subsets before pruning.

Input	Grid	n	M	$\binom{M}{k}$
Five groups	Hybrid, shifted	10^5	5	1
Five groups	Hybrid, origin zero	10^5	20	15,504
Five groups	Exchange, either origin	10^5	89,166–89,317	$\approx 4.7 \times 10^{22}$
Ten groups	Exchange	10^6	10	252
Uniform square	Exchange	10^6	644,670–645,259	$\approx 9.3 \times 10^{26}$
Swiss sites	Hybrid	5,403	706	$\approx 7.95 \times 10^{21}$

E.2 Paired certificate protocol

The paired study fixes $\varepsilon = 1/2$ and uses 114 configurations (99 distinct input arrays). Each is evaluated on PC, PK, I, X and H. Of 72 configurations per method eligible for subset search, the common limits are $M \le 160$, $\binom{M}{k} \le 10^7$, and five seconds. Across the five methods, 288 searches completed, 72 exceeded a size limit, and none timed out. The other 42 configurations per method (210 runs) measure occupancy only. Neither exclusions nor occupancy measurements produce an approximate solution.

Generators. For c groups, start at $(i \bmod w, \lfloor i/w \rfloor)$, $w = \lceil \sqrt{c} \rceil$. Irregular centers add independently sampled uniform offsets in $[-.3, .3]^2$. Group populations are balanced, except in the unequal

family, which uses one point per group plus a multinomial allocation with a Dirichlet$(1, \ldots, 1)$ probability vector. Within each group sample without replacement from a square lattice of step $4/2^{20}$ and half-width ρ. The elongated family stretches offsets by eight horizontally. Rotated and elongated inputs then rotate by .37 radians. Finally round to denominator 2^{20}, shuffle, and check distinctness and exact-integer bounds. Uniform controls sample the square $[0, \sqrt{c})^2$ and undergo the same final rounding and validation. This generator produces synthetic candidate groups, not a model fitted to cities.

The $k = 5$, $n = 2000$, $c = 10$ family varies $\rho \in \{1/4096, 1/1024, 1/256\}$ across six geometric types and three seeds. Nine additional origin controls use the same irregular inputs at $\rho = 1/1024$ and physical origins $o = (ah_X, ah_X)$, $a \in \{1/4, 1/2, 3/4\}$, held fixed across all methods. They preserve the input hash and all distances. The nine exact cases use $n = 24$, $k = 4$, $c = 8$, $\rho = 1/32$ and irregular, elongated, or uniform input. The 42 occupancy cases use $n = 10^5$, $c = 2k$, $\rho = 1/256$, $k \in \{8, 16, 32, 64, 128, 256, 512\}$, and lattice or irregular centers. Different k changes c in that study, so it does not isolate cardinality on an unchanged input.

Pairing and timing. Each method uses the same points and input order, the same traversal prefix, identical representative selection, common physical origin, and the same rational branch-and-bound routine. The dyadic scale and hence cell width depend on the certificate. Method execution order rotates with the seed. Shared traversal and incumbent-evaluation times are logged separately from method-specific grid and search time; data generation and validation are excluded from those stage times. Each input-method pair is timed once, so seed ranges are not confidence intervals. Timings were taken sequentially on an AMD Ryzen 7 5800H, Windows 11, with Python 3.12.2 and NumPy 1.26.4. At fixed dimension the grid stage uses $O(n)$ storage. The optional exact solver stores pair weights on the representatives; the separate Swiss audit also materializes full-input squared distances. These implementations do not operate on a stream.

Reproduction and validation. Work in `anc/benchmarks/`. The recorded compact data are sufficient to regenerate Figure 2, Table 7 and the sweep summary with `python code/report.py`. To rerun the measurements, use `python code/scale.py`, `python code/paired.py` and `python code/validation.py`, then regenerate the summaries. The raw runs include ordered retained indices; the compact records retain their hashes and counts, the grid widths, exact energies when available, and timings. The two search drivers resume by configuration ID. Sixteen small inputs with independently enumerated optima support the certificate and local-optimum checks in `code/checks.py`; together with input reproducibility checks this gives 157 checks.

Interpretation. PC and PK agree for small k because pair count is then the tighter analytic bound. At large k exchange can improve on the evaluated incumbent in both center families, yet the resulting sets can still be too large for enumeration. Across the 72 cases with a search protocol X retained fewer/equal/more sites than I in $0/13/59$; among the 42 cases measuring occupancy alone the split was $12/9/21$. H attained the smaller count in every case by grid nesting. The scale study above and the paired study use different generators and budgets; we report them separately.

Table 7: All paired configurations at $\varepsilon = 1/2$. Each row aggregates seeds $17, 29, 43$. Entries are median representative counts; Cardinality rows ($n = 10^5$) measure occupancy only. o/h_X is the shared physical origin relative to the exchange width.

Family	ρ	n	k	o/h_X	PC	PK	I	X	H
lattice	$1/4096$	2000	5	0	40	40	40	40	40
lattice	$1/1024$	2000	5	0	40	40	40	40	40

Family	ρ	n	k	o/h_X	PC	PK	I	X	H
lattice	$1/256$	2000	5	0	40	40	40	43	40
irregular	$1/4096$	2000	5	0	12	12	12	16	12
irregular	$1/1024$	2000	5	0	18	18	16	40	16
irregular	$1/256$	2000	5	0	40	40	24	229	24
rotated	$1/4096$	2000	5	0	11	11	11	16	11
rotated	$1/1024$	2000	5	0	15	15	13	43	13
rotated	$1/256$	2000	5	0	43	43	25	237	25
unequal	$1/4096$	2000	5	0	12	12	12	16	12
unequal	$1/1024$	2000	5	0	18	18	14	38	14
unequal	$1/256$	2000	5	0	38	38	23	208	23
elongated	$1/4096$	2000	5	0	16	16	14	43	14
elongated	$1/1024$	2000	5	0	43	43	26	179	26
elongated	$1/256$	2000	5	0	177	177	77	1094	77
uniform	$1/4096$	2000	5	0	1948	1948	1948	1998	1948
uniform	$1/1024$	2000	5	0	1948	1948	1948	1998	1948
uniform	$1/256$	2000	5	0	1948	1948	1948	1998	1948
lattice	$1/256$	100000	8	0	115	115	64	115	64
irregular	$1/256$	100000	8	0	143	143	64	399	64
lattice	$1/256$	100000	16	0	595	595	198	595	198
irregular	$1/256$	100000	16	0	793	793	286	793	286
lattice	$1/256$	100000	32	0	4194	4194	340	1119	340
irregular	$1/256$	100000	32	0	5077	5077	572	1582	572
lattice	$1/256$	100000	64	0	31340	31340	2140	2140	2140
irregular	$1/256$	100000	64	0	32955	32955	3118	9964	3118
lattice	$1/256$	100000	128	0	83251	51398	16452	4201	4201
irregular	$1/256$	100000	128	0	83527	52989	19166	19166	19166
lattice	$1/256$	100000	256	0	97657	91133	31349	8288	8288
irregular	$1/256$	100000	256	0	97692	91294	34796	34796	34796
lattice	$1/256$	100000	512	0	99704	95533	83390	16458	16458
irregular	$1/256$	100000	512	0	99701	95511	84091	54751	54751
irregular	$1/1024$	2000	5	$1/4$	19	19	14	38	14
irregular	$1/1024$	2000	5	$1/2$	16	16	14	38	14
irregular	$1/1024$	2000	5	$3/4$	15	15	13	40	13
irregular	$1/32$	24	4	0	23	23	23	24	23
elongated	$1/32$	24	4	0	24	24	24	24	24
uniform	$1/32$	24	4	0	24	24	24	24	24

E.3 Varying cardinality on fixed inputs

Does a coarser exchange grid at large k translate into a completed solve? We used six fixed arrays and varied only k within each. The first panel samples 2,048 points in 24 tight irregular groups (half-width $1/4096$); the second samples 1,024 points around 256 lattice centers (half-width $1/256$). Each panel uses seeds 17, 29 and 43, accuracy $1/2$, the common physical origin $(0, 0)$ and the same deterministic first-index representative. Coordinates and their order stay fixed. For each of the 57 array/cardinality pairs we built the incumbent (I), exchange (X) and hybrid (H) grids, rotating method order. The primary search required $M \leq 160$ and $\binom{M}{k} \leq 10^7$, with a separate five-second wall-clock budget per admitted solve, including reduced pair-weight construction and excluding worker startup and grid building.

The only strict exchange improvements in this fixed-input study occurred on all three lattice arrays at $k = 128$. In seed order the counts (M_I, M_X) were $(1016, 958)$, $(998, 933)$ and $(1004, 924)$.

Table 8: Fixed-input cardinality study at $\varepsilon = 1/2$. Each M is the median over three seeds. The last column in each panel counts completed primary searches out of three for I/X/H. A dash denotes an untested cardinality. A size exclusion gives an occupancy count but no reduced optimum. All admitted primary searches completed.

	24 irregular groups ($n = 2048$)				256 lattice groups ($n = 1024$)			
k	M_I	M_X	M_H	completed	M_I	M_X	M_H	completed
4	25	31	25	3/3/3	698	698	698	0/0/0
5	27	31	27	3/3/3			—	
8	27	39	27	3/0/3	698	698	698	0/0/0
10	27	39	27	2/0/2			—	
11	27	39	27	1/0/1			—	
12	27	39	27	1/0/1	698	698	698	0/0/0
13	31	2048	31	0/0/0	698	698	698	0/0/0
16	31	2048	31	0/0/0	698	698	698	0/0/0
32			—		698	698	698	0/0/0
64			—		933	933	933	0/0/0
96			—		933	933	933	0/0/0
128			—		1004	933	933	0/0/0
192			—		1004	1024	1004	0/0/0
256			—		1020	1024	1020	0/0/0

To test pruning directly, we launched each I/X/H solve for that cardinality without the two size gates. All nine reached the five-second limit. This says what the specified implementation accomplished under that budget; it is not a lower bound on search time. These experiments do not certify an optimum when search fails.

From `anc/benchmarks/`, run `python code/fixed_inputs.py --check` to regenerate the six arrays, compare them with their frozen integer coordinates, rebuild every grid and verify exact energies of the returned subsets. Use `--summary` for the recorded table or `--run` to repeat the timed experiment in a fresh copy. The point and run records are `results/fixed-input-points.json` and `results/fixed-input-runs.json`; they include input and representative hashes, certificate values, grid widths, search sizes and statuses.

F Staircases, nearly collinear sets, and cutoff proxies

These alternatives illustrate the difference between an exact structural parameter and an error certificate. We describe the rational planar case $s = 2$. The first two approaches exploit an order on most sites; the third exploits a sparse lower bound on a dense objective.

On a coordinatewise monotone staircase in Manhattan distance, reflect coordinates as needed and order the sites along the staircase. The coordinate $u = x + y$ then satisfies $\|z_j - z_i\|_1 = u_j - u_i$ for $i < j$. If deleting p exceptional sites leaves such a staircase, find a maximum staircase by longest-chain dynamic programming, then enumerate which exceptions are selected. For each choice their interactions with a core site become a unary cost. This adds only linear terms to the threshold minimum-cut formulation on a line [9], so the optimum is exact in $O(2^p n^6)$ arithmetic operations. The same argument applies to an exactly collinear Euclidean core, found by testing lines through pairs of sites.

For a nearly collinear Euclidean core, suppose instead that a strip of width W and rational axis direction is supplied, with p exceptional sites. Project only core–core interactions onto the axis and assign infinite cost to projected collisions. Keep all interactions involving exceptions

exact. Once the selected exceptions are fixed, sites sharing one projected coordinate have the same projected interactions with the rest of the core; among them, keep the least expensive unary choice. The ordered solver allows at most one selected site per projected coordinate and takes $O(2^p n^6)$ arithmetic operations. If the projected optimum V is finite, let U be the original energy of a selection attaining V. Then

$$\frac{V}{1+W^2V} \;\le\; \mathrm{OPT} \;\le\; U \;\le\; V.$$

Indeed if the projected pair cost is a, the true cost is at least $a/(1+W^2a)$ by Pythagoras; an unchanged term also satisfies this inequality. Summing gives a lower bound of $Q/(1+W^2Q)$ for a selection of projected cost Q, by using the common denominator $1+W^2Q$. A projected collision alone costs at least W^{-2} in the original metric when $W>0$. If $V=+\infty$, every feasible selection has such a collision, so W^{-2} is a lower bound and any feasible selection supplies an upper bound. At $W=0$ the projection is exact.

A radius $R>0$ can encode a modeling decision that sites farther apart are sufficiently separated. Retain their inverse-square interaction only for pairs at distance at most R, and let G_R be the graph of those pairs. Given a tree decomposition of width t, storing the selected bag vertices and the selected count gives an exact proxy optimum in $2^{O(t)}n(k+1)^2$ operations after polynomial preprocessing. For completeness, use a nice decomposition with an introduce-edge node for each retained interaction. A state (X,j) fixes the selected vertices X in the bag and counts j selected vertices already forgotten. Introducing a vertex branches on its membership; introducing an edge charges its weight if both endpoints are in X. On forgetting a selected vertex increment j. At joins use matching X and convolve the two forgotten counts: the charged edges and forgotten vertices of the children are disjoint. At an empty root read the minimum at $(\varnothing,k)$. For a proxy minimizer S_R, its proxy value is a lower bound on the full optimum and $E_2(S_R)$ an upper bound. Their difference is exactly its omitted tail, at most $\binom{k}{2}R^{-2}$. This certifies additive error even when the proxy value is zero; a useful relative guarantee needs a positive lower bound large enough compared with that tail.

To strengthen the lower endpoint in Section 6, take distance-bin endpoints $0<R_1<\cdots<R_m=D$ with D at least the input diameter. A pair in bin $(R_{j-1},R_j]$ ($R_0=0$) receives weight R_j^{-2}, at most its true cost. Subtract D^{-2} from each pair weight and form a graph of the strictly positive residuals. The last bin creates no edges. Adding $\binom{k}{2}D^{-2}$ to the exact minimum residual cost gives a lower bound B on the original optimum. The same treewidth dynamic program applies with the supplied decomposition; sharper bins may increase its width. No such proxy was needed for the independently verified Swiss optimum.

G Reproducing the computations

The PCGA-Riesz repository contains the implementation, Swiss site coordinates and exact solution records. The archived experimental release (`arxiv-v1`) also provides the ordered benchmark point arrays, the full measurement logs and SHA-256 checksums. Its `points-index.csv` maps each of the 339 recorded cases to an array, coordinate denominator and checksum. Synthetic benchmark coordinates are integer numerators divided by 2^{20}. The Swiss CSV gives integer metres in LV95 (EPSG:2056).

Replay the reported results. With Python 3.12 and NumPy 1.26.4, run the following command from the root of either the repository or the extracted arXiv source archive:

```
python anc/reproduce.py --swiss
```

The driver works offline on a temporary copy. It rebuilds the benchmark summaries, checks all 171 grids on the six fixed inputs, verifies the returned subset energies, and repeats the small geometric examples. The Swiss audit recomputes the two reduced optima, the full-input optimum and the sequence of eight accepted hybrid moves. Allow about 1 GB of free memory. Omit `-swiss` for the shorter replay. Recorded timeouts remain observations from the original machine.

Run an individual study. The paths in Appendices B–E refer to the arXiv source archive. To recreate that layout from a Git checkout, run

```
python anc/reproduce.py --export ../riesz-paper
```

from the repository root, choosing a new directory for the export. For example, in `anc/benchmarks/` of the exported copy, `python code/fixed_inputs.py --check` rebuilds the grids in Table 8. The benchmark README gives the seeds, budgets and commands for fresh timings. Run timed experiments sequentially. Exact scores and cell membership are deterministic for a completed solve.

To typeset the paper, keep `main_arxiv_v1.tex` and `PRIMEarxiv.sty` together and run pdfLaTeX until the references settle. All figures and bibliographic entries are inline.

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